

DevOps for Databases & Data Warehouses: Deliver Changes Faster, Safely and Reliably

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CEO, MakeStuffGo



- 25+ years mucking about with 1s & 0s
- Bringing DevOps to databases (and the masses) is a personal passion
- Understanding data and cloud is a company driver
- Technologist who understands business value...
- #makeStuffGo



@theHybridDBA



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Before we start...

- Make sure you have VS Code (free)
 - + the mssql extension
 - + the SQL Database Projects extension
- A (free) GitHub account
- .NET 8 SDK
- A database somewhere

Before we start...

- MAIN LAB:

<https://sqlworkshop.makestuffgo.com/>

- OPTIONAL LAB:

<https://letsmakestuffgo.github.io/database-devops-autopilot/>

What you signed up for...

This workshop is a deep dive on **database DevOps centered on the SQL database projects format**, where you will learn practical techniques for managing database changes whether your workload is operational, analytical, or somewhere in between. Our exploration begins with the foundational CI/CD capabilities for SQL projects, where you can ensure no matter how a **database is developed it is in source control** and can be verified before deployments to one or more environments. The same shared SQL projects format applies to the **whole Microsoft SQL family**, so we'll examine the advantages and special capabilities to understand when implementing DevOps practices for Fabric SQL, Azure SQL, or SQL Server. We conclude our workshop with some special topics at the core of **good data DevOps practices**, including security management, static data management, and coordinating changes with other workloads.

Ground rules

- Follow the conference code of conduct:
 - “Harassment-free conference experience”
 - “Deliberate intimidation”
 - “Sustained disruption”
- Your questions are welcome
 - No bad questions
 - May have to hold on a question due to schedule
- Try to leave space for all the knowledge in the room
- Practice the human side of DevOps - empathetic, collaborative

Pre-Conference Session Agenda / Timing

The agenda/timing for Pre-Con sessions on Nov 17-18 is as follows:

START TIME	END TIME	STATUS
09:00 AM	10:45 AM	Pre-Cons in session
10:45 AM	11:00 AM	Refreshment break
11:00 AM	12:30 PM	Pre-Cons in session
12:30 PM	1:30 PM	Lunch
1:30 PM	3:15 PM	Pre-Cons in session
3:15 PM	3:30 PM	Refreshment break
3:30 PM	5:00 PM	Pre-Cons in session
5:00 PM	–	END OF DAY
TOTAL	8:00	6.5 HOURS OF CONTENT

Agenda

9am

Intro

Foundations of SQL
projects

10am

SQL Variants

Advanced Topics Part
One

12pm

 Lunch 

2pm

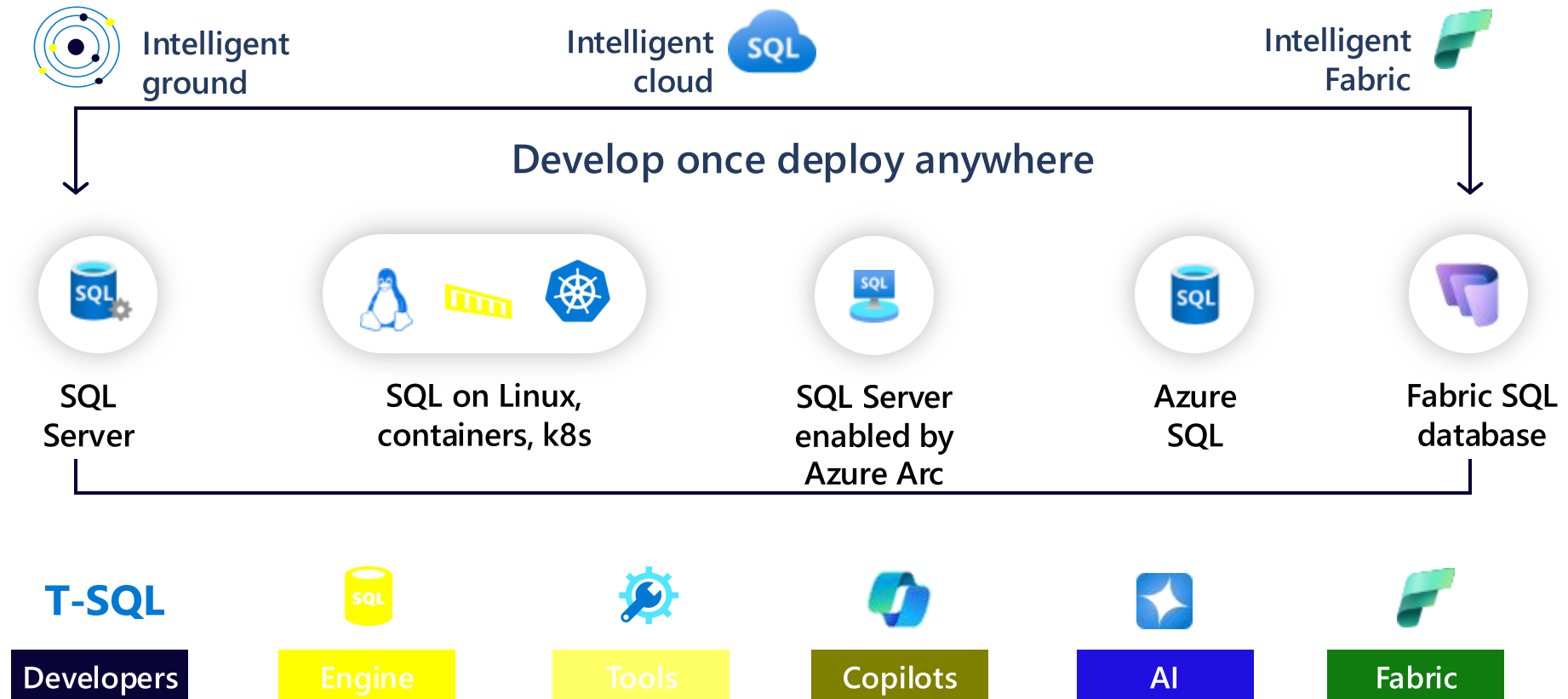
Advanced Topics Part
Two

Troubleshooting and
Optimizing DevOps
Workflows

5pm

Closing

Microsoft SQL - a whole range



Why DevOps?

- **Issues identified as early as possible (build)**
- **Scalability and Flexibility**
 - **Add team members and know the releases will be consistent**
- **Consistent releases**
- **Increased Efficiency**
- **Removes single point of failure (person)**
- **Automated documentation**
- **Improved quality**

*Team looks like left side of this tree
, flourishing, achieving more and more,
people want to be in the team*



Why Not DevOps?

- Errors identified at production time
- Team unclear on how to deploy
- ~~Legacy Systems~~
- Small/Simple projects
- Resistance to change
- Regulatory Constraints

*Team ends up looking
like this tree,
unmotivated, confused*

What is DevOps?



Tools



Process



People



Culture

There are a few tools out there.....

1 Fm Gh Github	PERIODIC TABLE OF DEVOPS TOOLS (V2) EMBED DOWNLOAD ADD Os Open Source Fr Free Fm Freemium Pd Paid En Enterprise SCM CI Deployment Cloud / IaaS / PaaS BI / Monitoring Database Mgmt Repo Mgmt Config / Provisioning Release Mgmt Logging Build Testing Containerization Collaboration Security
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XebiaLabs
Deliver Faster

 Follow @xebialabs

91 En Xlr XL Release	92 En Ur UrbanCode Release	93 En Bm BMC Release Process	94 En Hp HP Codar	95 En Au Automic	96 En Pl Plutora Release	97 En Sr Serena Release	98 Pd Tfs Team Foundation	99 Fm Tr Trello	100 Pd Jr Jira	101 Fm Rf HipChat	102 Fm Sl Slack	103 Fm Fd Flowdock	104 Pd Pv Pivotal Tracker	105 En Sn ServiceNow
106 Os Ki Kibana	107 Fm Nr New Relic	108 Os Ni Nagios	109 Os Zb Zabbix	110 En Dd Datadog	111 Os El Elasticsearch	112 Os Ss StackState	113 En Sp Splunk	114 Fm Le Logentries	115 Fm Sl Sumo Logic	116 Os Ls Logstash	117 Os Gr Graylog	118 Os Sn Snort	119 Os Tr Tripwire	120 En Ff Fortify

DevSecOps Tools Periodic Table

AiOps Artifact/Package Management Collaboration Configuration Automation Container Orchestration Continuous Integration Database Management Deployment Enterprise Agile Planning IT Service Management PaaS/Container Service Public Cloud Release Management Security Source Control Management Testing Value Stream Management Developer Portal DevOps AI-ML Analytics										1 En Aja Atlassian Jira Align 3 En Daa Digital.ai Agility 4 En Tp Targetprocess 11 En Pv Planview 12 En Br Broadcom Rally										2 Os Gi Git 5 En Azp Azure DevOps Pipelines 6 Os Ow OWASP ZAP 7 En Dap Digital.ai App Protection 8 En Ck CyberArk Conjur 9 En Sw ServiceNow 10 Os Gh GitHub 13 En Dad Digital.ai Deploy 14 En Sni Sonatype Nexus IQ 15 En Aq Aqua Security 16 En Vc Veracode 17 En Bi BMC Helix ITSM 18 Os Gls GitLab SCM									
Aj En Atlassian Jira Dd En Datadog Bp En Big Panda In Pd Instana Acp En AWS CodePipeline Mt En Microsoft Teams Rha Os Red Hat Ansible Ht En HashiCorp Terraform Dk Os Docker Rho En Red Hat OpenShift Lb Os Liquibase Dp Fm Delphix Ud En UrbanCode Deploy Om Pd OpxMx Hv Os HashiCorp Vault Sy En Snyk Pd Os PagerDuty Abb Fm Atlassian Bitbucket										37 En Sp Splunk Ad En AppDynamics Kb En Kibana Dar En Digital.ai Release Ur En UrbanCode Release Ac En Atlassian Confluence Ch Os Chef Acf En AWS Cloud Formation Ku Os Kubernetes Ak En Amazon EKS De En Docker Enterprise Rf En Redgate Flyway Ha En Harness Pi En Pulumi Sr Os SonarQube Ff En Micro Focus Fortify SCA Azf En Azure Functions Ci En Compuware ISPW										Dt En Dynatrace Nr En New Relic Dh En Docker Hub Np En npm Ja En JFrog Artifactory So En Stack Overflow Sl En Slack Hc Os HashiCorp Consul Pu Fm Puppet Azk En Azure AKS Ae En Amazon ECS Qt Fm Quest Road Sk Os Spinnaker Od En Octopus Deploy Sb En Synopsys Black Duck Cx En Checkmarx SAST He Fm Heroku Al En AWS Lambda									
Gr Os Grafana El Os Elastic ELK Stack Yn Os Yarn Nu Os NuGet Snx Os Sonatype Nexus Mm Os Mattermost Mr En Miro MI En Mural Hp Os HashiCorp Packer Gk En Google GKE Hm Os Helm Fx Os Flux Tk Os Tekton Acd Os AWS CodeDeploy Sn Os Snort Pbs Fm PortSwigger Burp Suite Gf En Google Firebase Cf Os Cloud Foundry										91 Os Jn Jenkins Azc En Azure DevOps Code Glc Os GitLab CI Tr Os Travis CI Cc Fm Circle CI Mv Os Maven Ab Pd Atlassian Bamboo Ga En Github Actions Acb En AWS CodeBuild Cf En CodeFresh Az En Azure Gc En Google Cloud Aws En AWS Os Os OpenStack Bg Os Backstage										Tt Fr Tricentis Tosca Se Fr Selenium Ju Fr JUnit Sl Pd Sauce Labs Ct En Compuware Topaz Ap En Appium Sq Os Squash TM Cu Fr Cucumber Jm Fr Jmeter Pa Pd Parasoft Dac En Digital.ai Continuous Testing Da En Digital.ai Pvz En Planview Viz Pr En Plutora Dai En Digital.ai Intelligence									

What does success look like when working with SQL and DevOps?

- Hamish talking points
 - Reliable Database Deployments
 - Environment Consistency
 - Foundation of Monitoring & Observability
 - Auditors like DevOps
 - Faster Development & Deployment Cycles
 - SAFER Development & Deployment Cycles
 - Apps -> Infrastructure -> Databases -> Analytics
 - Developers, Data Engineers, DBAs collaborate
 - Governance and compliance are less boring

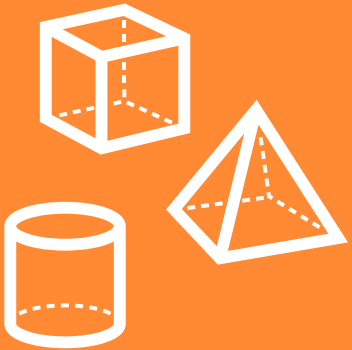
Database as code – SQL projects



Database settings



Pre-deployment script



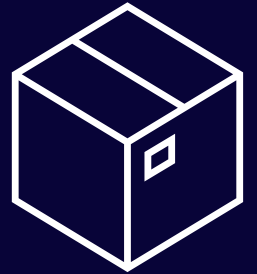
Database objects
(tables, stored procs, indexes, etc.)



Post-deployment script

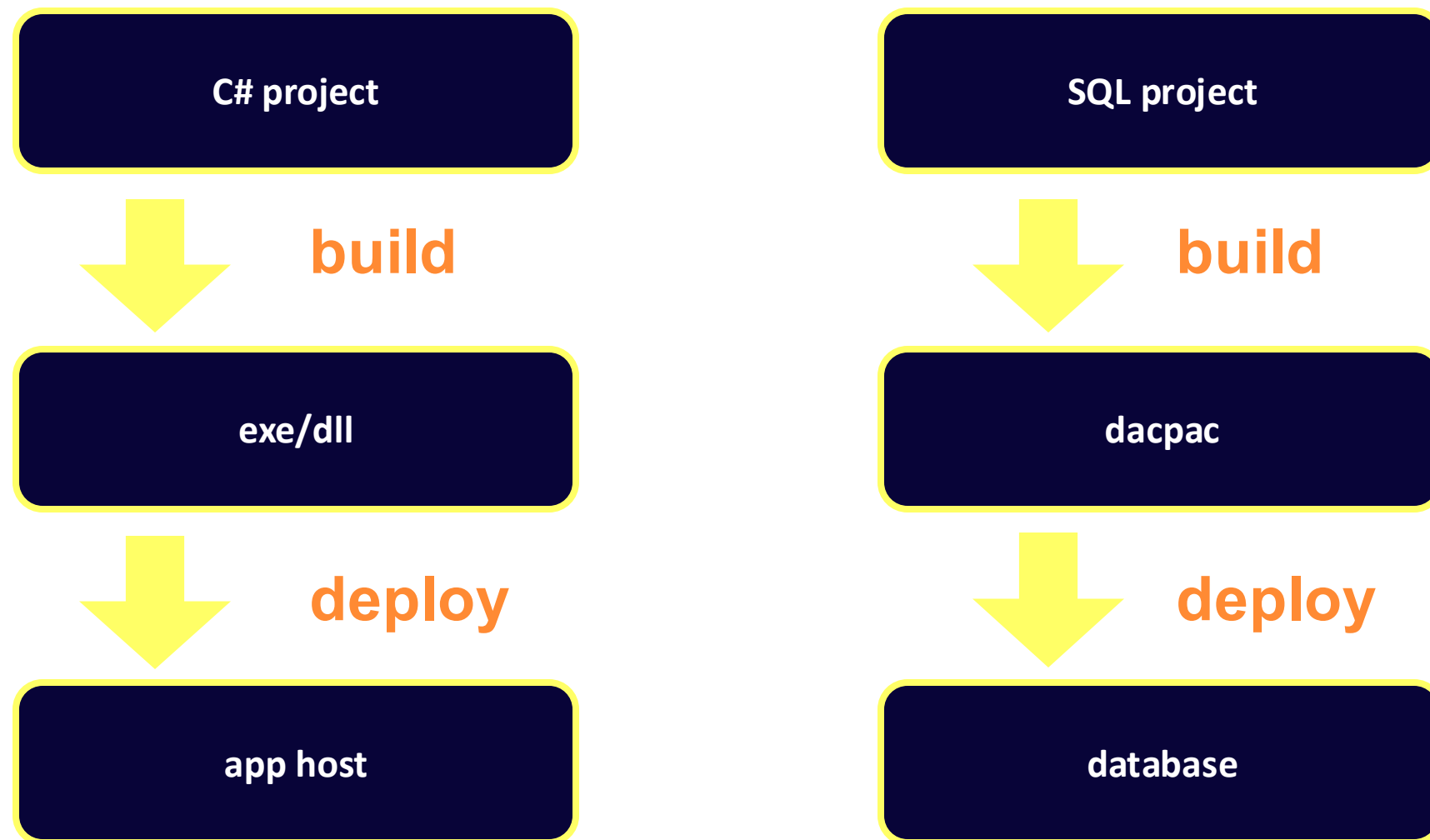
build

.dacpac

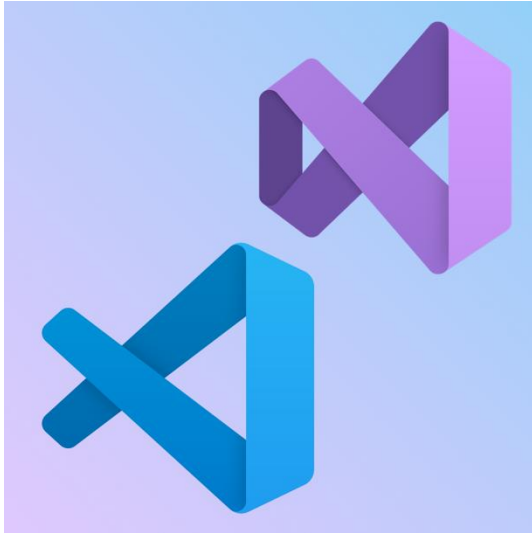


Foundations of CI/CD and SQL projects

Build and ship - project lifecycle

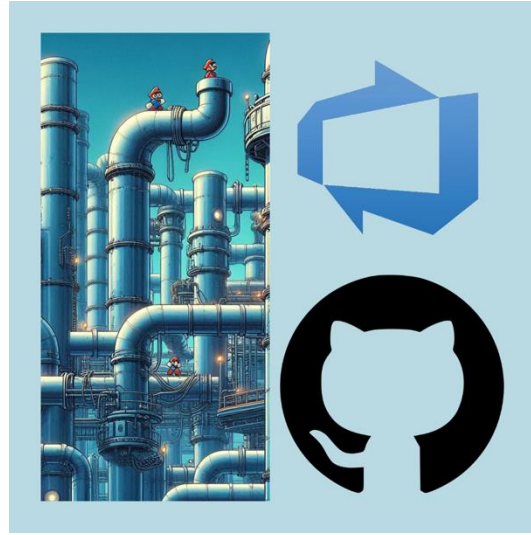


SQL projects tooling ecosystem



Visual Studio and VS Code

Develop, analyze, and compare database objects



GitHub and Azure DevOps tasks

Streamline SQL project deployment from CI/CD environments



SqlPackage CLI

Automate transposing between files and databases

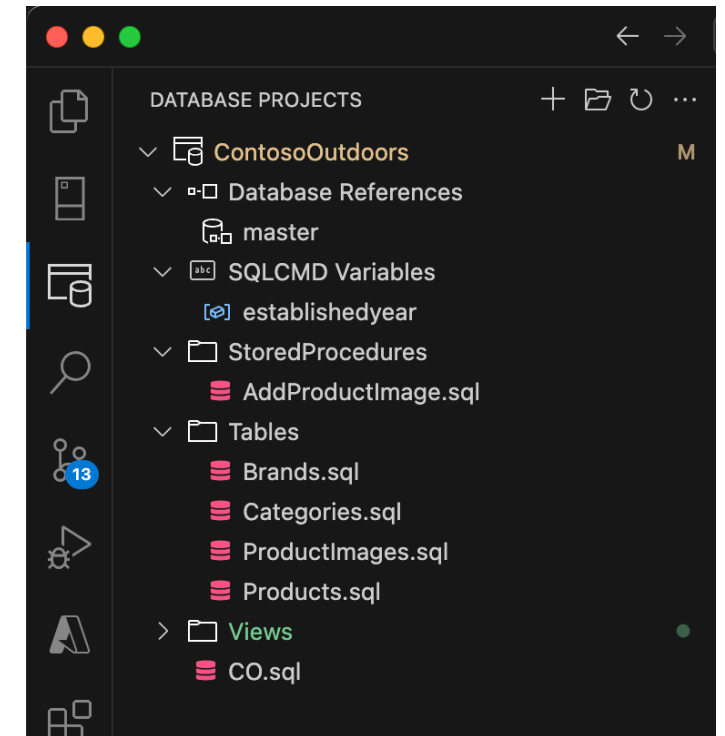
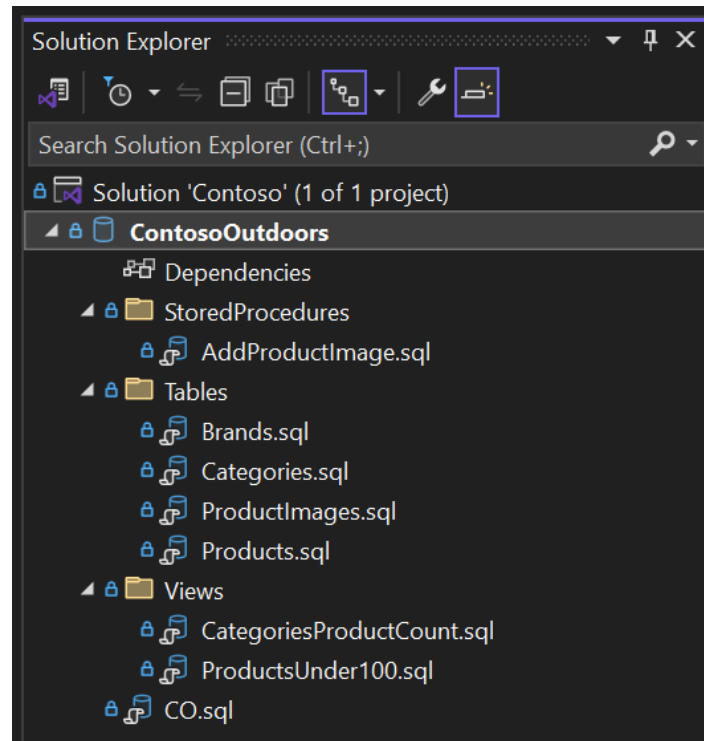
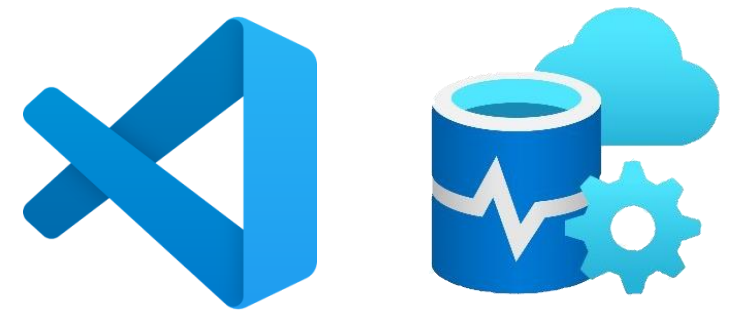
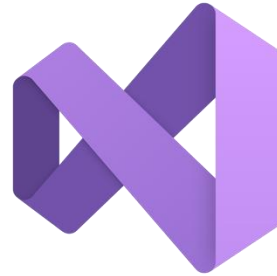


API, class, and model packages

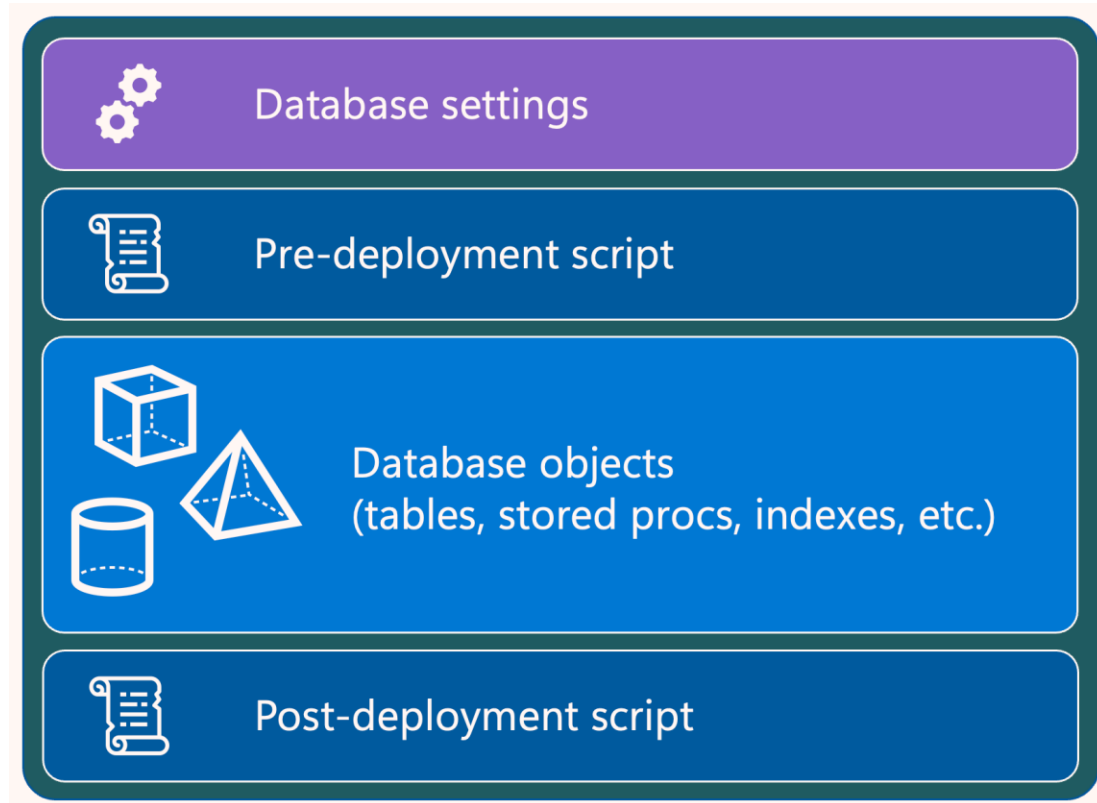
Extensibility for core APIs includes loading analysis rules and database parts from packages

Desktop tooling

- Add objects to projects
- Update projects from databases
- Modify project settings



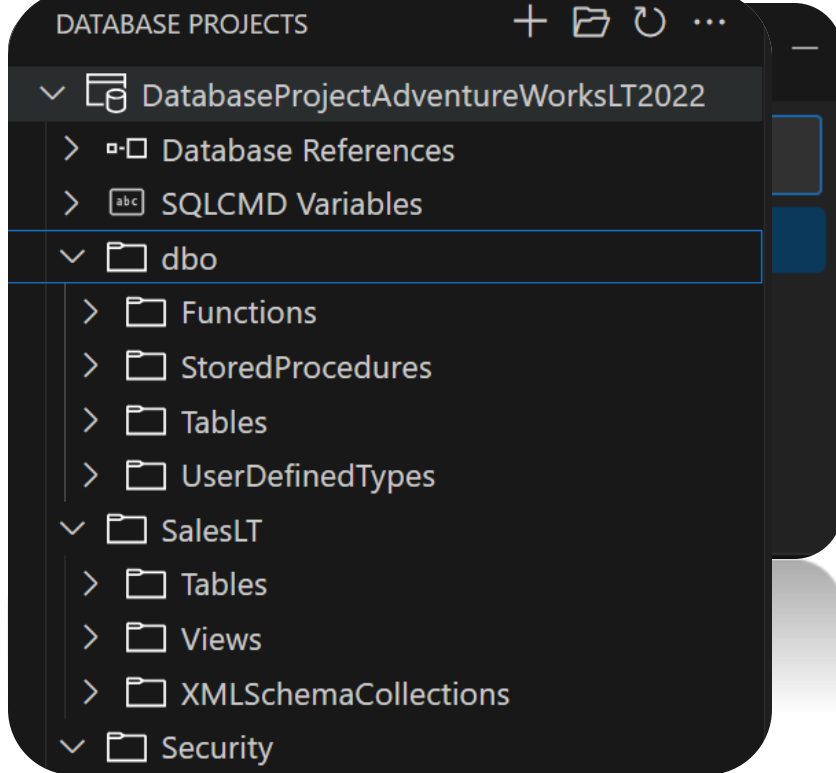
The .sqlproj file behind the scenes



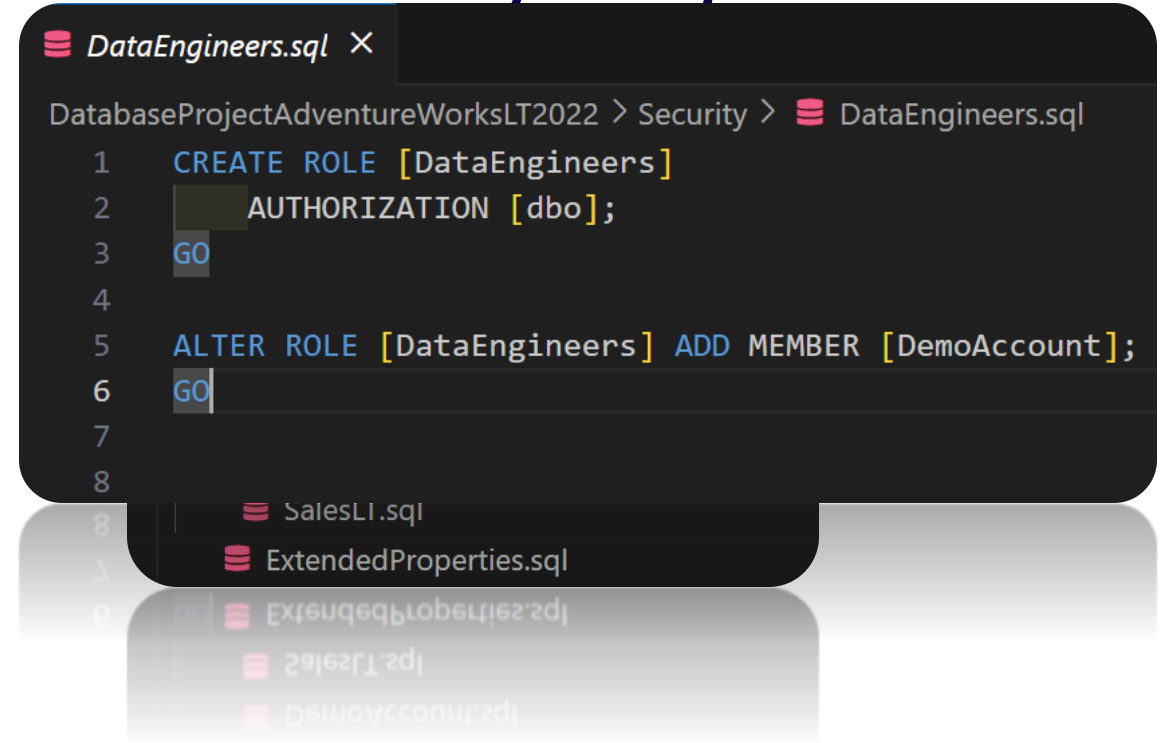
- Automatically includes all *.sql* files in the folders as objects
- Logins/users/roles/permissions == security objects
- You can store non-object scripts in same folders as `<None />` items
- Set specific scripts to execute on every deployment as `<PreDeploy />` and `<PostDeploy />` items
- Project properties establish database settings

Considerations for .sqlproj

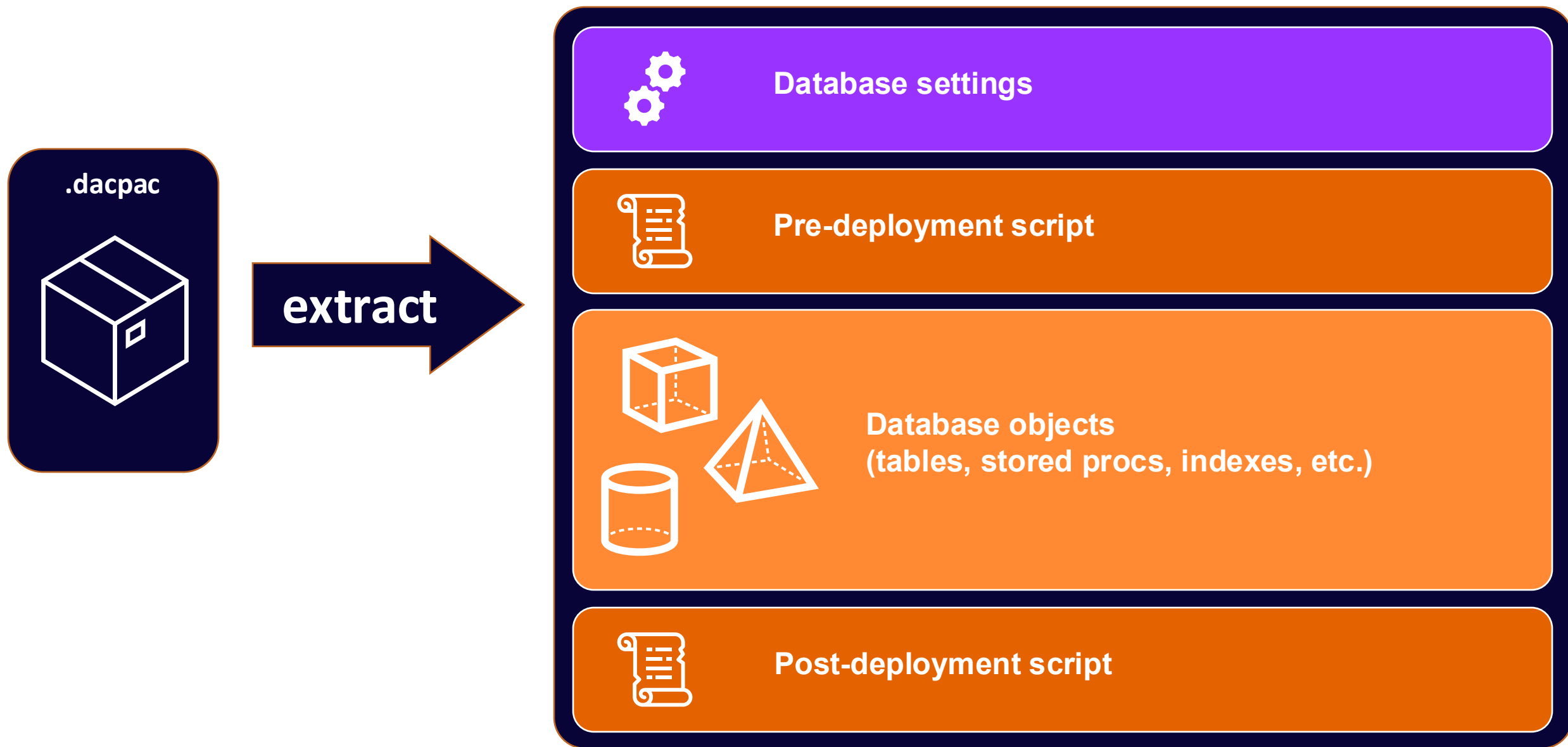
- Folder structure



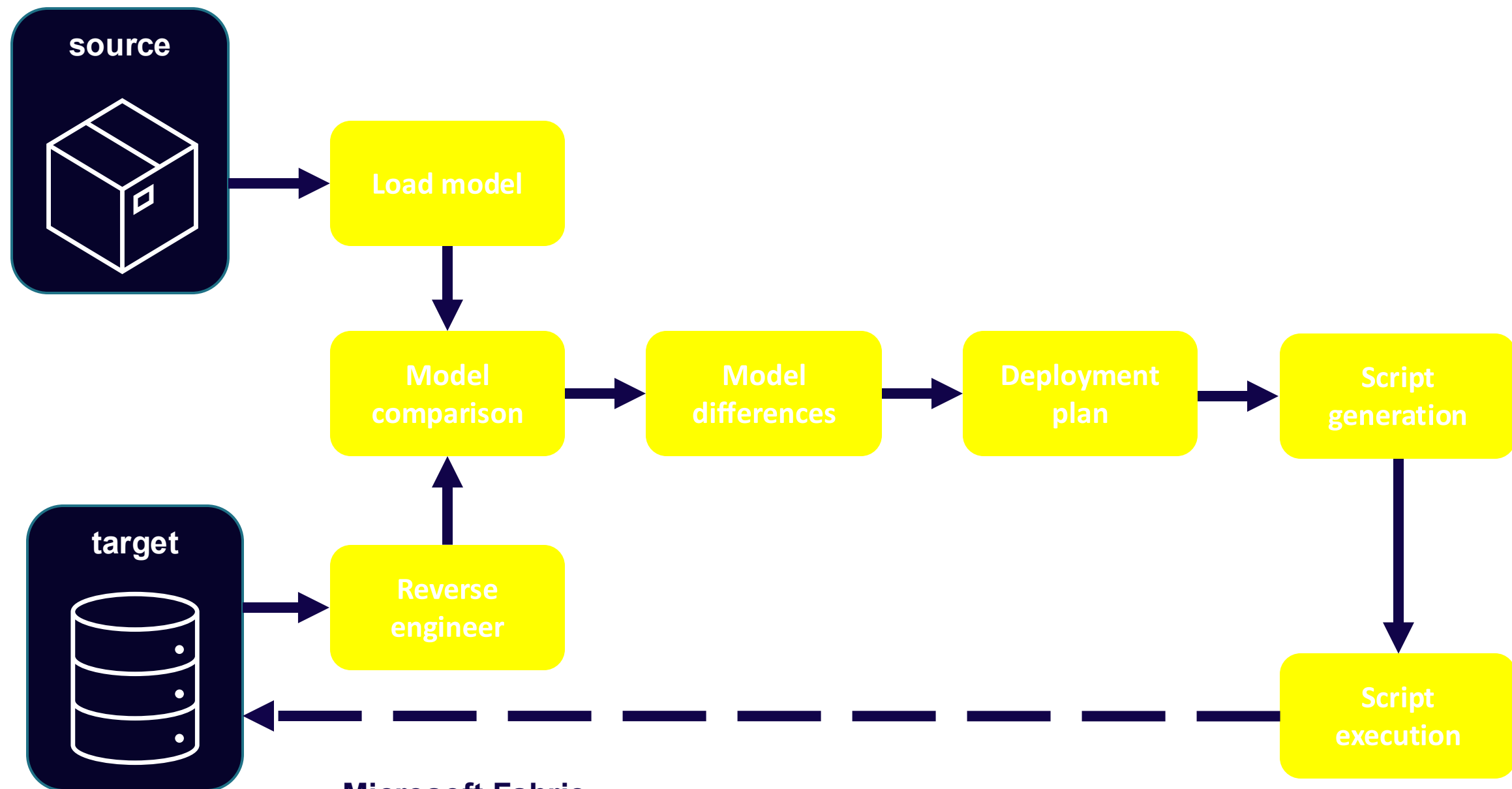
- Security objects



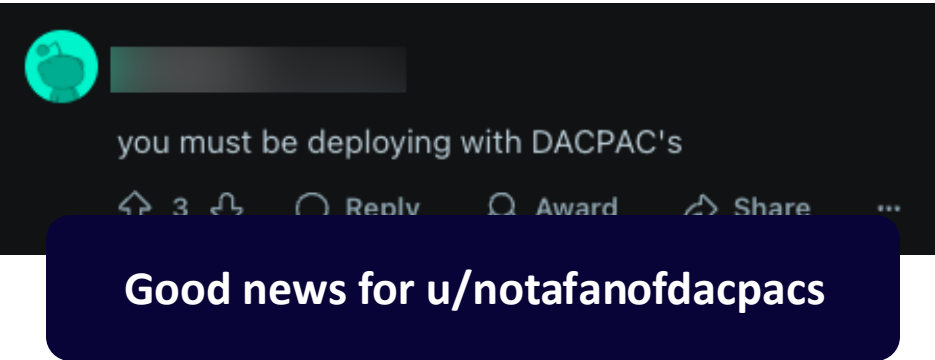
SqlPackage extract/publish



Dacpac deployment behind the scenes



Deployment surprises



De-risk SqlPackage deployments with:

1. Script (outputs .sql) for review
2. Deploy report, enabling automatic actions

CI/CD tooling

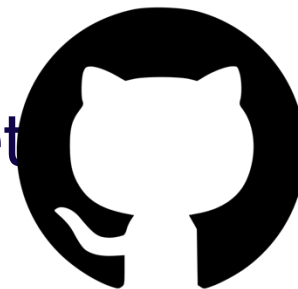
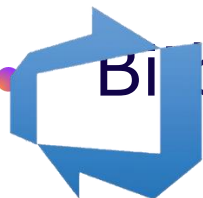
CLIs

- **SqlPackage**
- **Sqlcmd**
- ---
- Dbup
- App frameworks (EF Core)



Runtime hosts

- **GitHub actions**
- **Azure DevOps pipelines**
- ---
- Gitlab
- Bitbucket

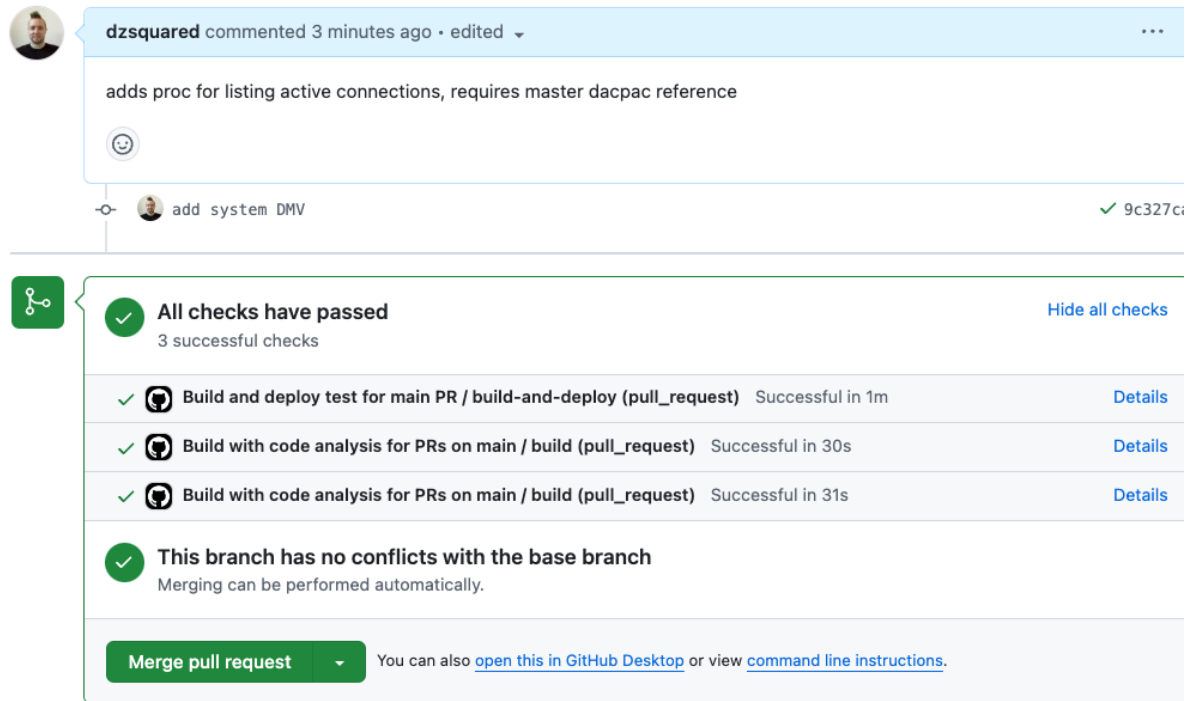


Managed environments

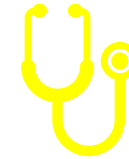
- **Fabric**



Continuous integration (CI)



Is it valid?



Is it good?



Is it correct?

- Leveraging automation in development processes to improve quality during collaboration
- Goal: knowing if this is checked in, it meets our standards

SQL code analysis

- “out of the box” option with SQL projects includes 14 rules
 - Design
 - Naming
 - performance
- When enabled, it runs during build
 - Project property (in .sqlproj)
`<RunSqlCodeAnalysis>True</RunSqlCodeAnalysis>`
 - Build-time property
`dotnet build /p:RunSqlCodeAnalysis=true`



Extending rules

Code analysis rules are written in C#

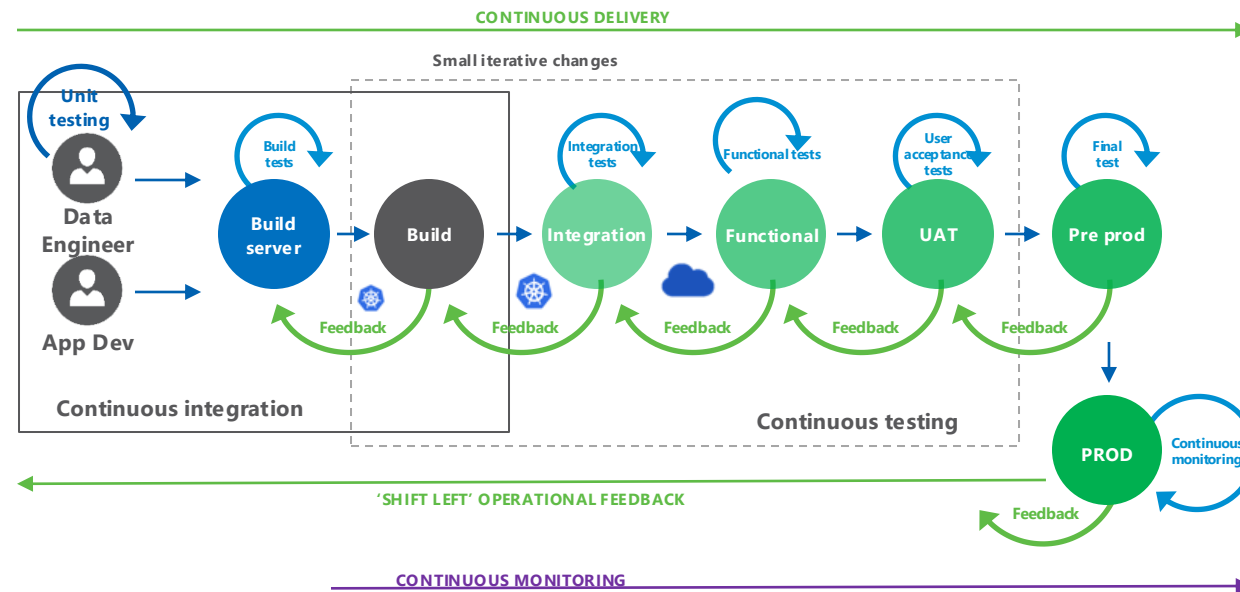
Rules have access to database model (including relationships)

Code analysis rule sets have been shared in the SQL community

Continuous deployment (CD)

- Deployment to where? (not necessarily prod)
- When code is checked in, it is deployed to *some environment* as soon as possible
- Goal: checked in code provides value quickly

CI/CD – create change continuously



How much git do you need to know?

Commit

Mark your current files as a point in the git timeline



Push

Send the local contents to a remote (like GitHub or Azure DevOps)
Branch-specific



Branch

Partition changes in different sections of the timeline
main
username/fix-1234
release/1_23



Merge/Pull

request
Combine
updates from one branch into another



Anatomy of a pipeline

GitHub: workflow
Azure DevOps: pipeline

YAML = yet another markup language

```
1  # This workflow will build a .NET project
2  # For more information see: https://docs.github.com/en/actions/automating-builds-and-tests/building-and-testing-net
3
4  name: Deploy SQL project
5
6  on:
7    workflow_dispatch:
8
9  jobs:
10    build:
11
12      runs-on: ubuntu-latest
13
14      steps:
15        - uses: actions/checkout@v4
16        - name: Setup .NET
17          uses: actions/setup-dotnet@v4
18          with:
19            dotnet-version: 8.0.x
20        - name: Restore dependencies
21          run: dotnet restore Wingtips/Wingtips.sqlproj
22        - name: Build
23          run: dotnet build Wingtips/Wingtips.sqlproj --no-restore
24        - name: Publish SQL project
25          run: |
26            SqlPackage /Action:Publish /SourceFile:Wingtips/bin/Debug/Wingtips.dacpac /TargetConnectionString:${{ secrets.SQL_CONNECTION_STRING }}
```

Trigger starts whole pipeline

Each job wraps steps + settings

Agent

Steps

Pipeline anatomy

Trigger

- On demand
- On file changes
- On pull requests
- ...

Agent

- Cloud hosted
 - Windows or Linux
 - Comes with preset software
 - You can install software during pipeline
 - Automatically updated every couple of weeks
- Self-hosted (you host)

Steps

- Anything you can script can be a step
- Pre-created “tasks” or “actions” combine logic

Tips for success in today's labs

- **Ask for help**
- Remember you're learning two things at once
 - YAML syntax – double check indentation and docs
 - DevOps principles - consider the logical order of steps
- Go one step at a time

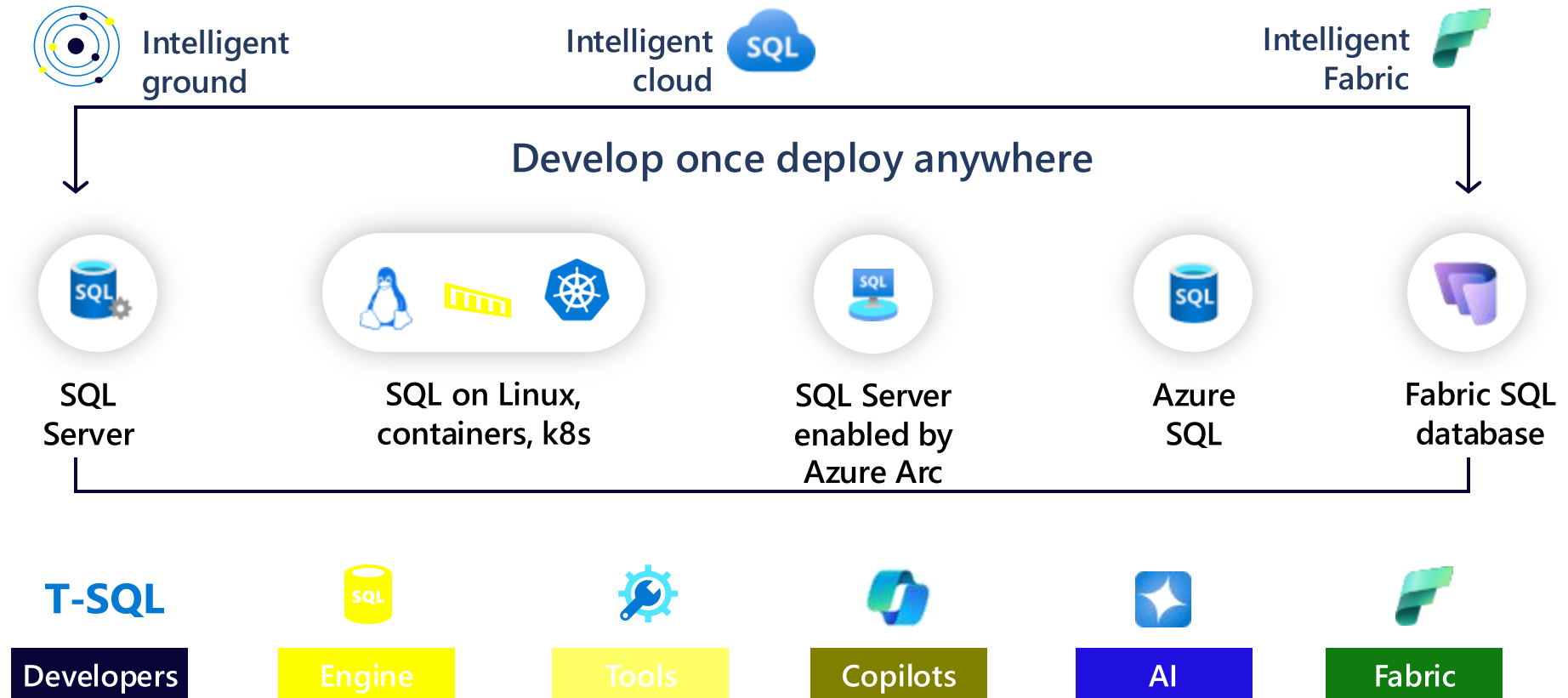
Lab 1 recap

1. Starting from an existing database, create a new project
2. Setup a publish pipeline such that new changes can be deployed (CD)
3. Setup a code analysis pipeline to help check new changes (CI)
4. Run a database change through the CI/CD

Lab One

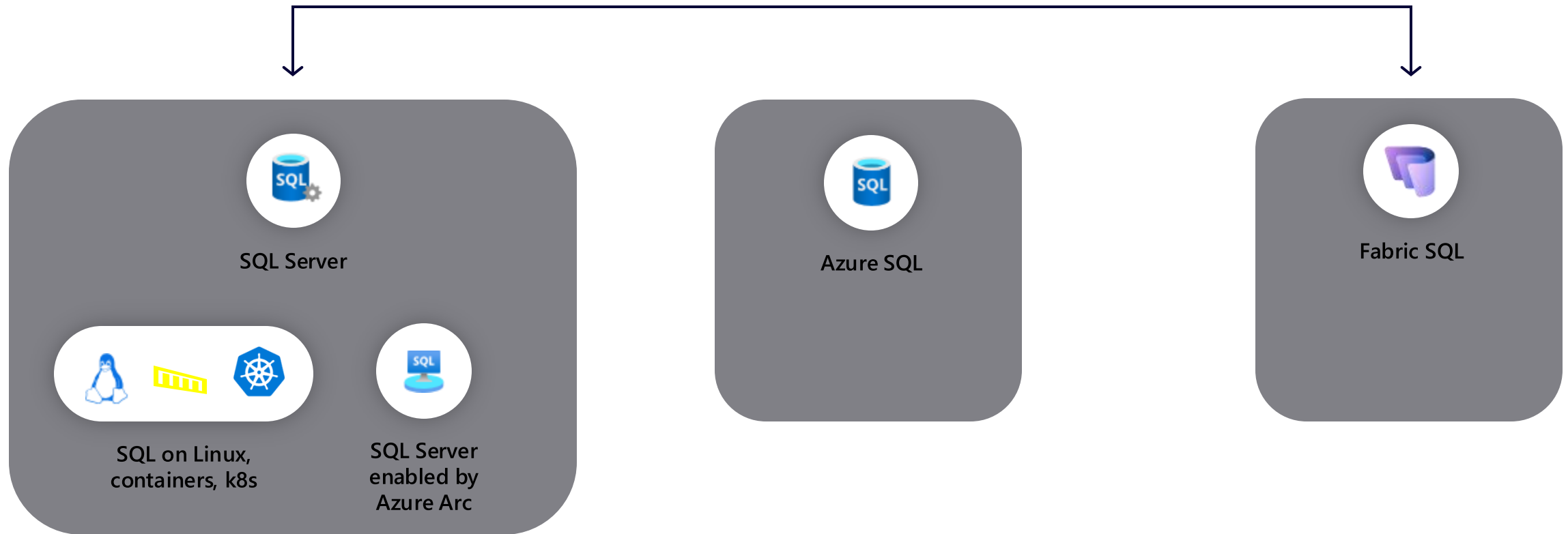
SQL projects across SQL variants

Microsoft SQL - a whole range



Grouping for SQL projects

Develop once deploy anywhere



Project properties

```
<?xml version="1.0" encoding="utf-8" ?>
```

```
<Project DefaultTargets="Build" >
```

```
  <Sdk Name="Microsoft.Build.Sql" Version="1.0.0" />
```

```
  <PropertyGroup>
```

```
    <Name>Wingtips</Name>
```

```
    <DSP>Microsoft.Data.Tools.Schema.Sql.SqlAzureV12DatabaseSchemaProvider</DSP>
```

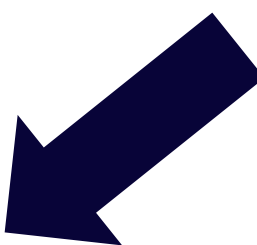
```
    <ModelCollation>1033,CI</ModelCollation>
```

```
    <ProjectGuid>{00000000-0000-0000-0000-000000000000}</ProjectGuid>
```

```
  </PropertyGroup>
```

```
</Project>
```

Controls what
features are
available



Change tracking, query store, ledger, collation, etc

Target platform

<DSP>

- What syntax is valid at build time?
- Guards against accidentally attempting to deploy a dacpac against a potentially incompatible engine
- Microsoft.Data.Tools.Schema.Sql.?
 - SqlAzureV12DatabaseSchemaProvider
 - Sql160DatabaseSchemaProvider
 - SqlDwDatabaseSchemaProvider
 - SqlDwUnifiedDatabaseSchemaProvider
 - SqlDbFabricDatabaseSchemaProvider
- Publish to a different target platform with SqlPackage
`/p:AllowIncompatiblePlatform=true`
- Build for a different target platform without updating the project file with
`dotnet build /p:DSP=dspvalue`

Curious about migration T-SQL compatibility?

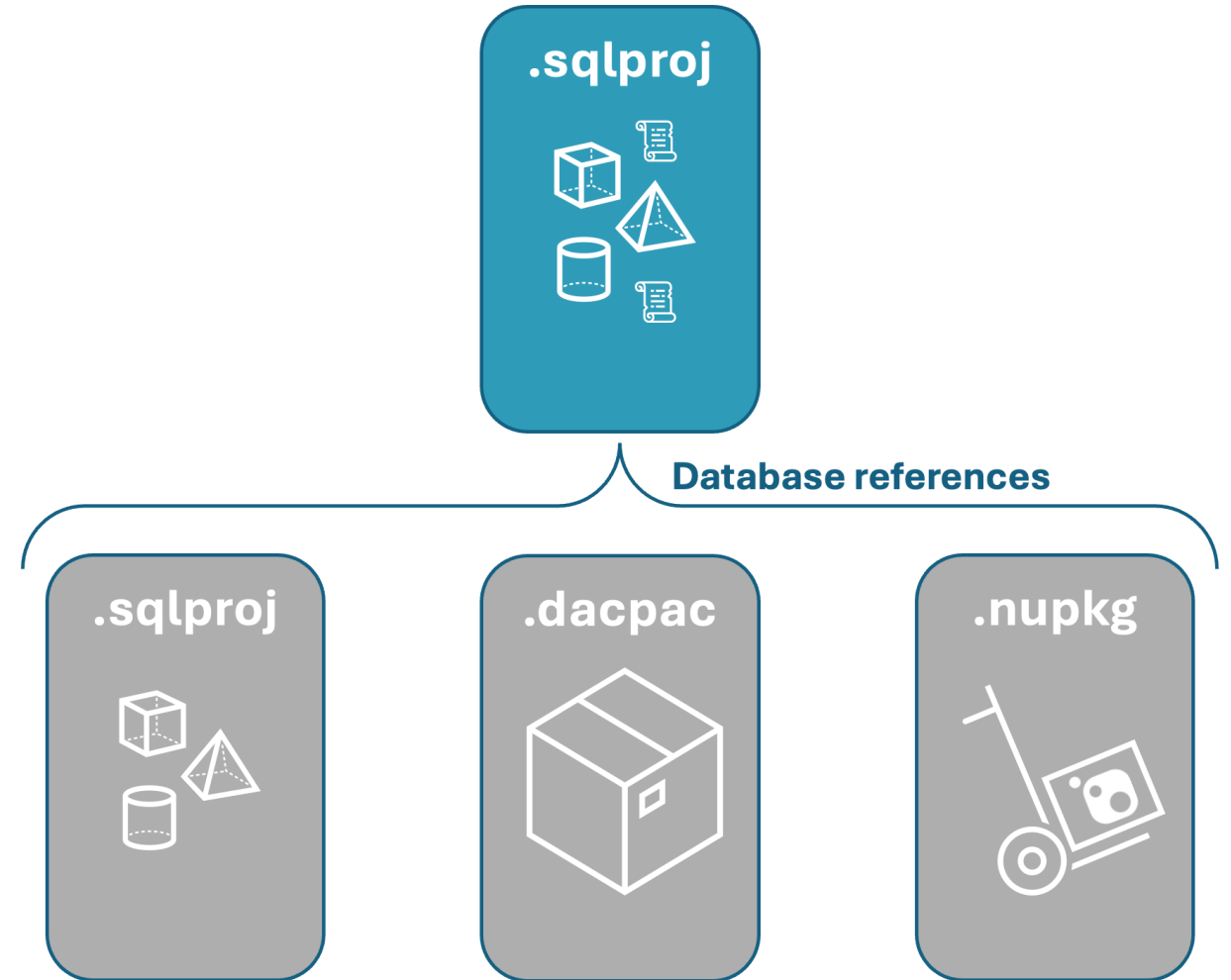


SQL Server

SQL projects resolve references to additional databases through

Database references

- Same database (more objects for that database) 📝
- Different database, same server (3-part naming) 🌳
- Different database, different server (4-part naming for linked servers) 💣



Database reference types

Project references

- All files must be present for every build
- Mono-repo or consolidated repos only
- Quickly iterate
- Longer build times for outer project

Dacpac references

- Dacpac file must be copied to build location (including in pipelines)
- Can be difficult to verify you have the right dacpac and consistently use the same path
- Since dacpac is already compiled, build times are improved

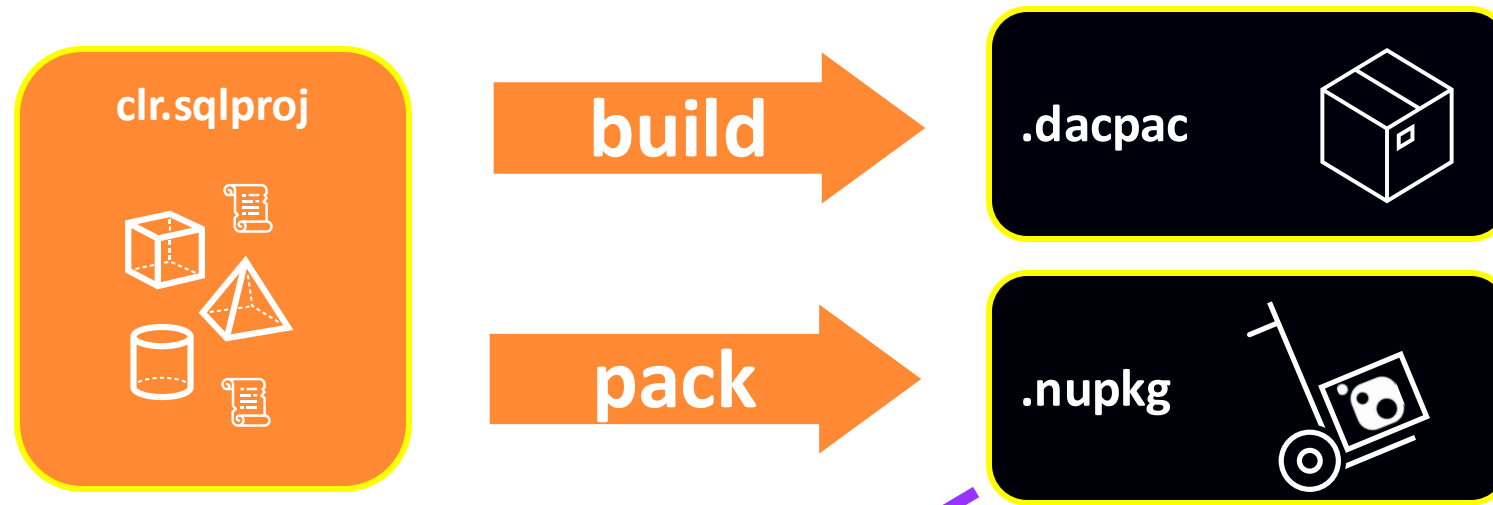
Package references

- “**dotnet pack**” creates a dacpac and packages it into a nupkg
- Referenced packages are automatically downloaded from the “package feed” where they were published
- Packages are referenced by version

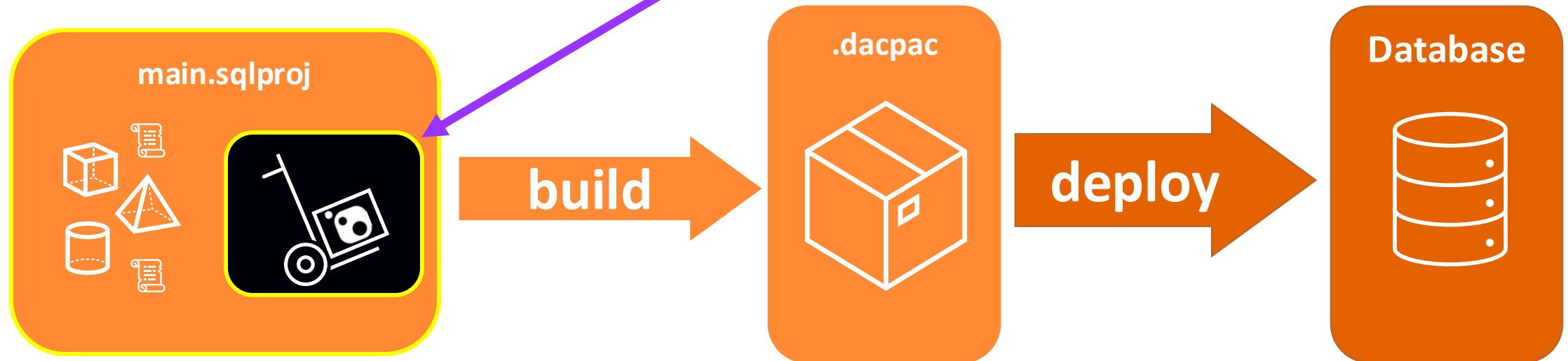
SQL Server - CLR objects

- SQL Server uses .NET Framework for CLR objects
- .NET Framework must be used to build the SQL project directly containing the CLR objects

.NET Framework



.NET



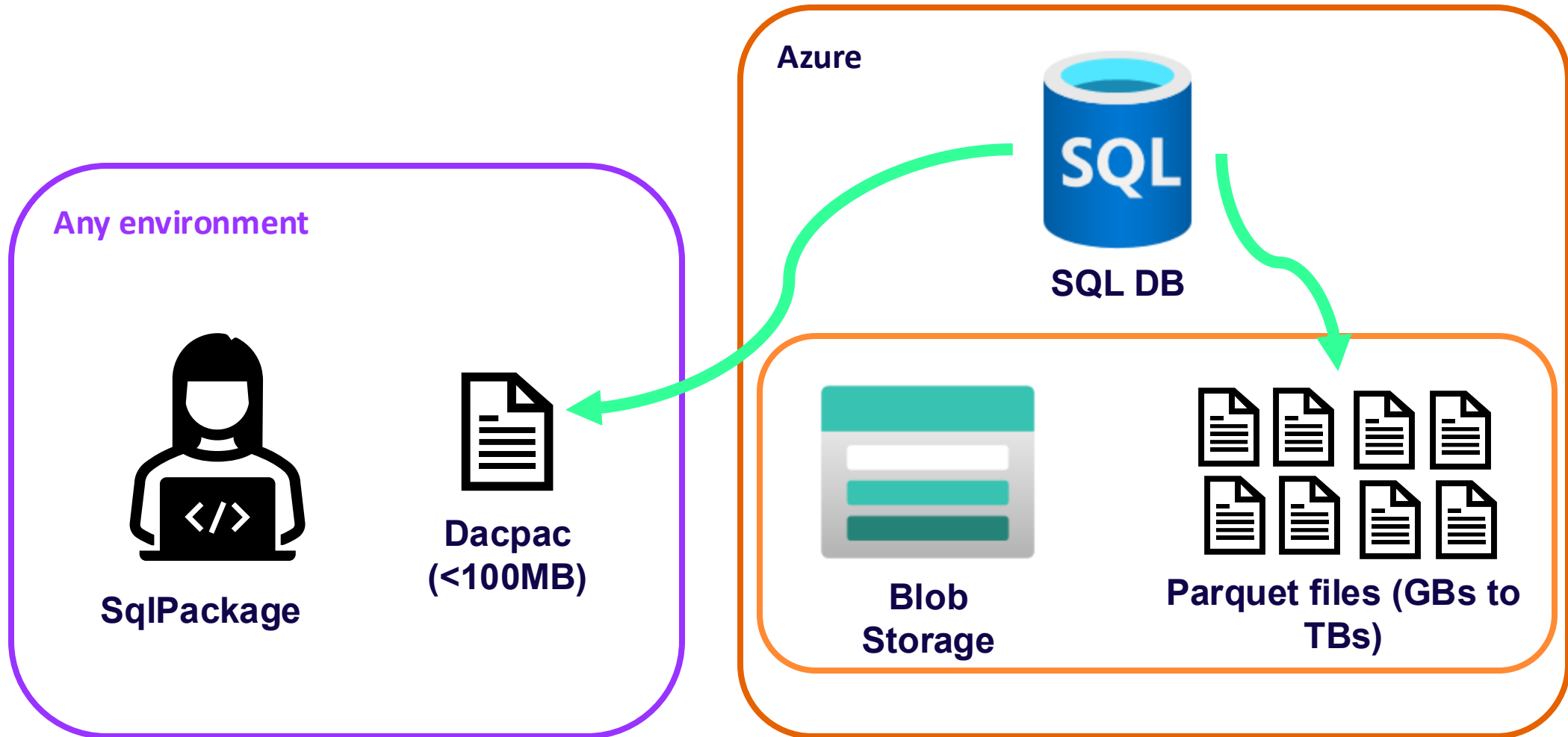
Azure SQL Database

- Reference master or system objects with master dacpac references
- 3-part naming not supported beyond system objects in the engine
- Database references can only be used for same-database references

Move data around with SqlPackage

- Bacpac contains data in contained bcp files by default
- Dacpac can include data in contained bcp files by default
- Dacpac can reference data in parquet files in Azure Blob Storage

Export (extract) to parquet in a fraction of the time



Azure SQL MI

“Always up to date” versionless

- Use Azure SQL DB target platform
- Allowed syntax will change

ov

Both support 3-part naming and same-server database references

Version-pinned Azure SQL MI

- Use SQL Server target platform
- Syntax is limited to capabilities of the SQL Server

pinned

Fabric and SQL projects

Data warehouse

- Analytics, big big data
- Tons of aggregation queries



SQL database in Fabric

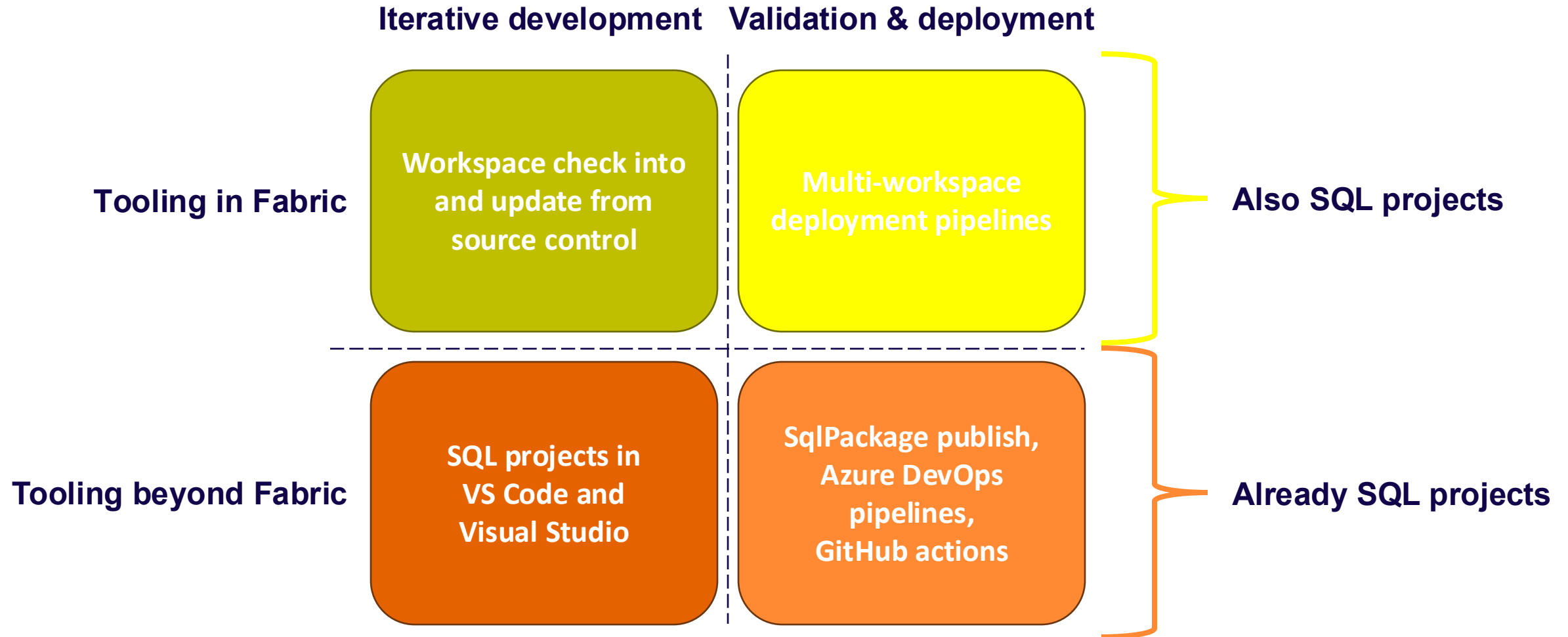
- OLTP, high performance CRUD
- Some analytical workloads
- (SQL Server, Azure SQL DB)
- No sql authentication



Lakehouse

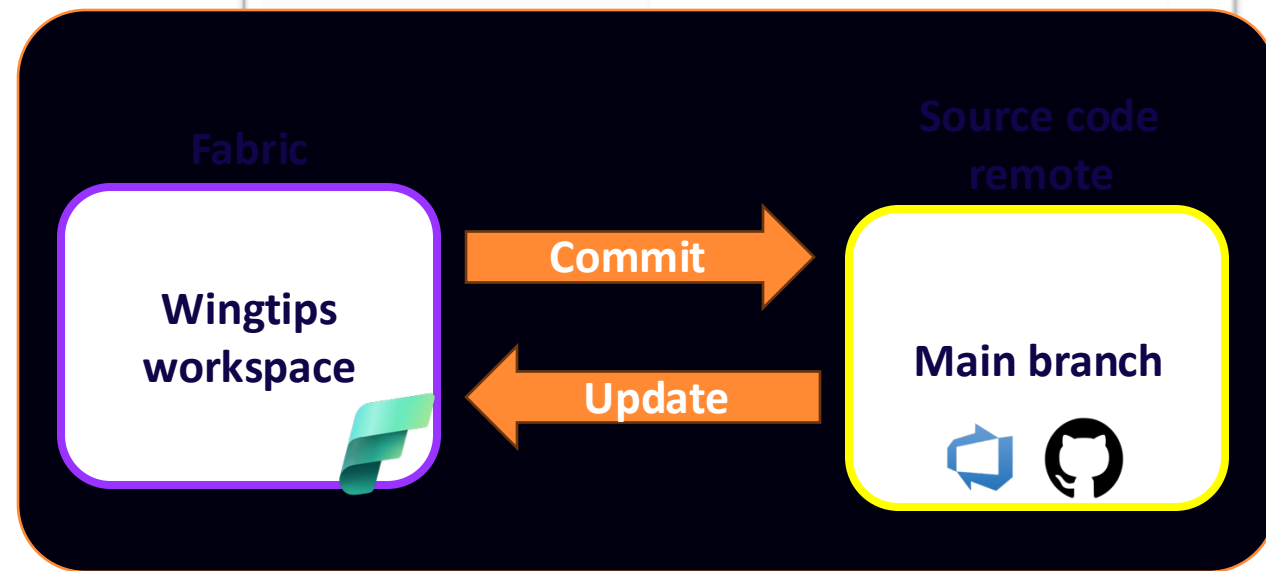
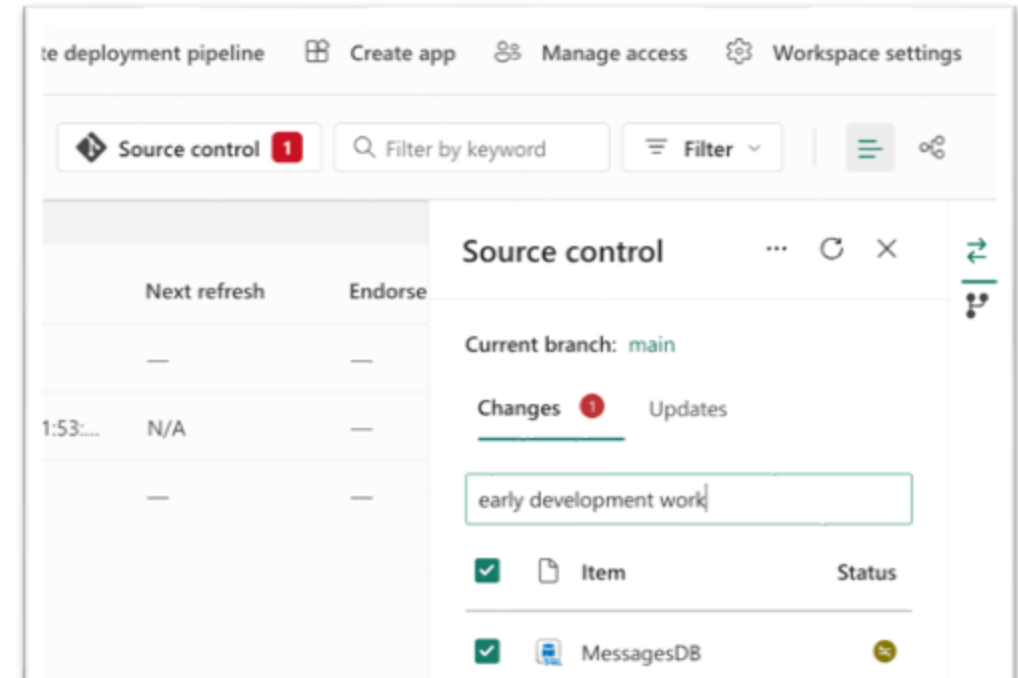
- (Views)

Fabric platform lifecycle management



Fabric source control integration

- GIT integration at workspace level
 - Commit workspace item definitions to source control
 - Update workspace items from source control definitions
- You can layer with deployment mechanisms
 - Fabric deployment pipelines
 - ADO Pipelines / GitHub actions



What's being written to source control?

The “item definition”

- .platform metadata file
- SQL project
- *Shared queries*

The screenshot shows a GitHub repository interface for 'ContosoLtd / FabricWorkspace / AdventureWorks.SQLDatabase / .platform'. The file explorer on the left shows the directory structure, with '.platform' selected. The main content area displays the JSON content of the file, which defines the schema and metadata for the AdventureWorks SQL database project.

```
<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Files
main + Q
Go to file t

> .github
✓ FabricWorkspace
✓ AdventureWorks.SQLDatabase
  > SalesLT
  > Security
  > dbo
    .gitignore
    .platform
    AdventureWorks.sqlproj
    Readme.md

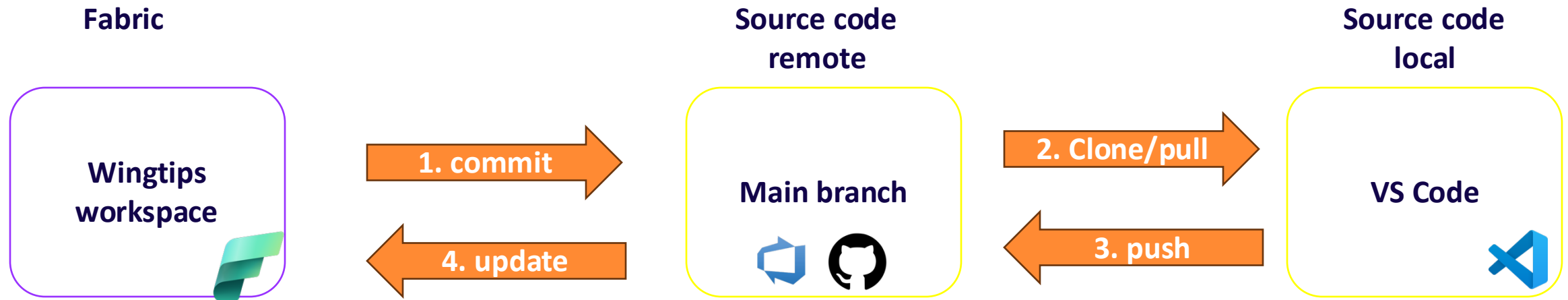
ContosoLtd / FabricWorkspace / AdventureWorks.SQLDatabase / .platform

dzsquared database setup 6b02e...

Code Blame 11 lines (11 loc) · 303 Bytes

1 {
2   "$schema": "https://developer.microsoft.com/json-schemas/fabric/gitInt
3   "metadata": {
4     "type": "SQLDatabase",
5     "displayName": "AdventureWorks"
6   },
7   "config": {
8     "version": "2.0",
9     "logicalId": "b0c513b8-afc6-b942-4707-d3b1de4a07ad"
10  }
11 }
```

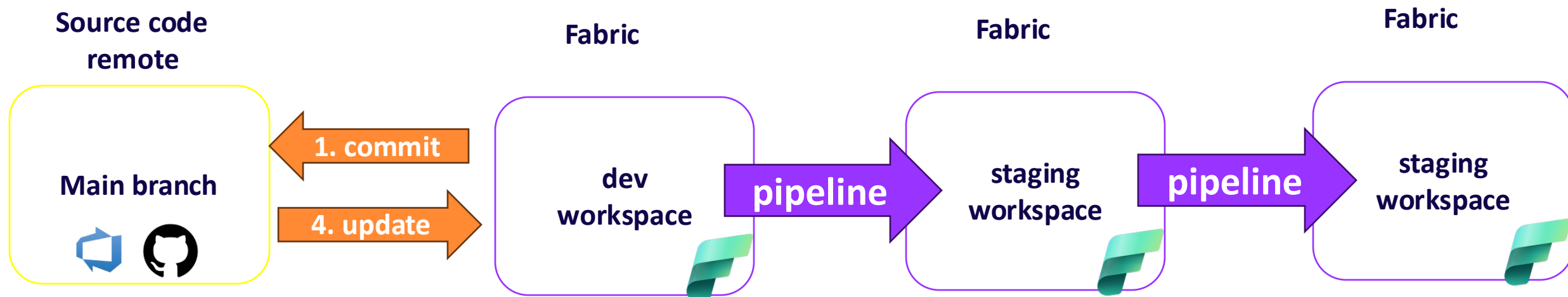
Leverage source control integration



	Name	Git status	Type	Task	Owner	Refreshed
	AdventureWorks	 Uncommitted	SQL database	—	Drew Skwiers-Kob...	—
	 AdventureWorks	—	Semantic model (...)	—	ContosoLtd	3/29/2025, 8:07
	 AdventureWorks	—	SQL analytics end...	—	Drew Skwiers-Kob...	—

Fabric deployment pipelines

- Alternative to GitHub actions, Azure DevOps pipelines
- Applies definitions from one workspace to another
- Limited customization and logging
- Quick to setup
- Provide object-level view of what has changed



	Name	Git status	Type	Task	Owner	Refreshed
	AdventureWorks	 Synced	SQL database	—	Drew Skwiers-Kob...	—
	 AdventureWorks	—	Semantic model (...)	—	ContosoLtd	3/29/2025, 8:07
	 AdventureWorks	—	SQL analytics end...	—	Drew Skwiers-Kob...	—

Microsoft Fabric

All your data. In one location. Organize. Collaborate. Create.

Explore the experience



Power BI
Find insights, track progress, and make decisions faster using rich visualizations.



Databases
Create operational SQL databases seamlessly for transactional workloads.



Data Factory
Solve complex data ingestion, transformation, and orchestration scenarios using cloud-scale data movement and data transformation services.



Industry Solutions
Use out-of-the-box industry data solutions and resources.



Real-Time Intelligence
Discover insights from your streaming data. Quickly ingest, index, and partition any data source or format, then query the data and create visualizations. You can also create alerts to flag anomalies.



Data Engineering
Create a lakehouse and operationalize your workflow to build, transform, and share your data estate.



Data Science
Unlock powerful insights using AI and machine learning technology.



Data Warehouse
Scale up your insights by storing and analyzing data in a secure SQL warehouse. Benefit from top-tier performance at petabyte scale in an open-data format.

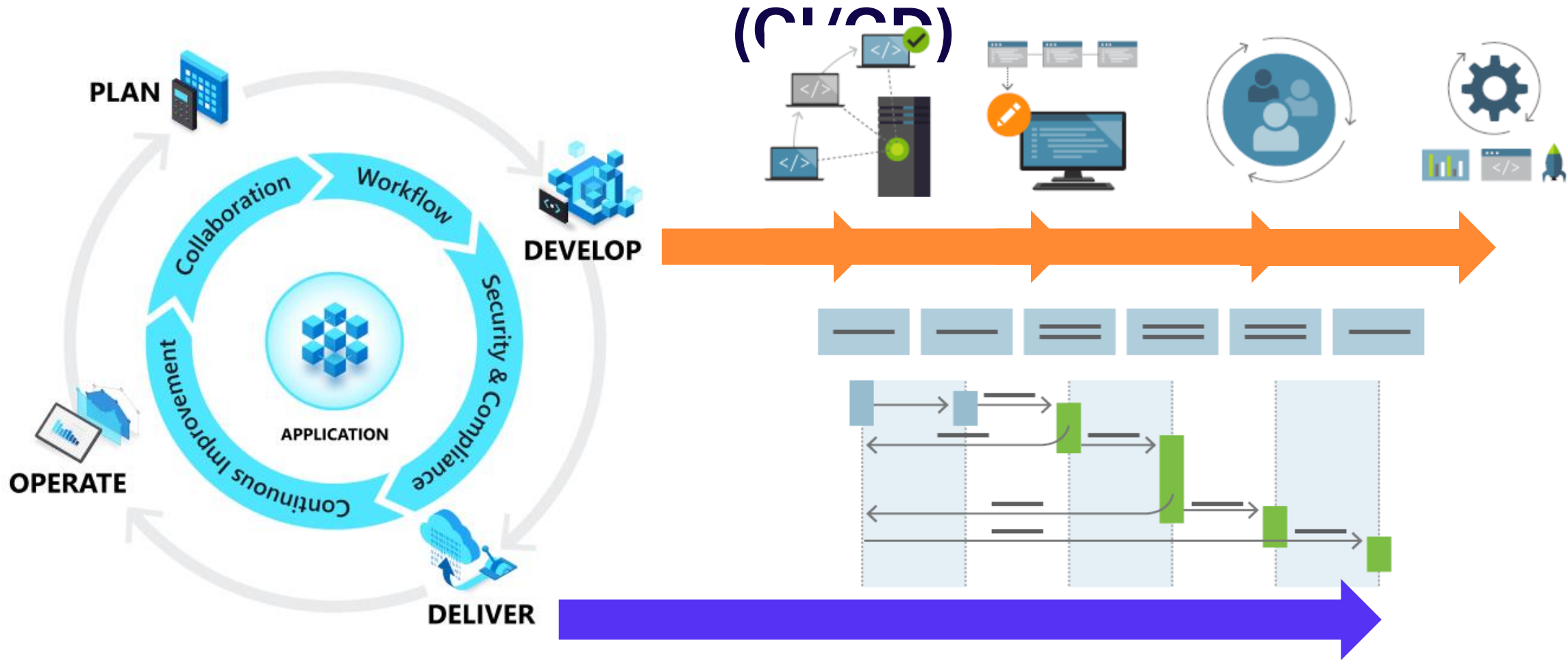


Read documentation



Explore community

Introduction to Continuous Integration & Continuous Delivery (CI/CD)



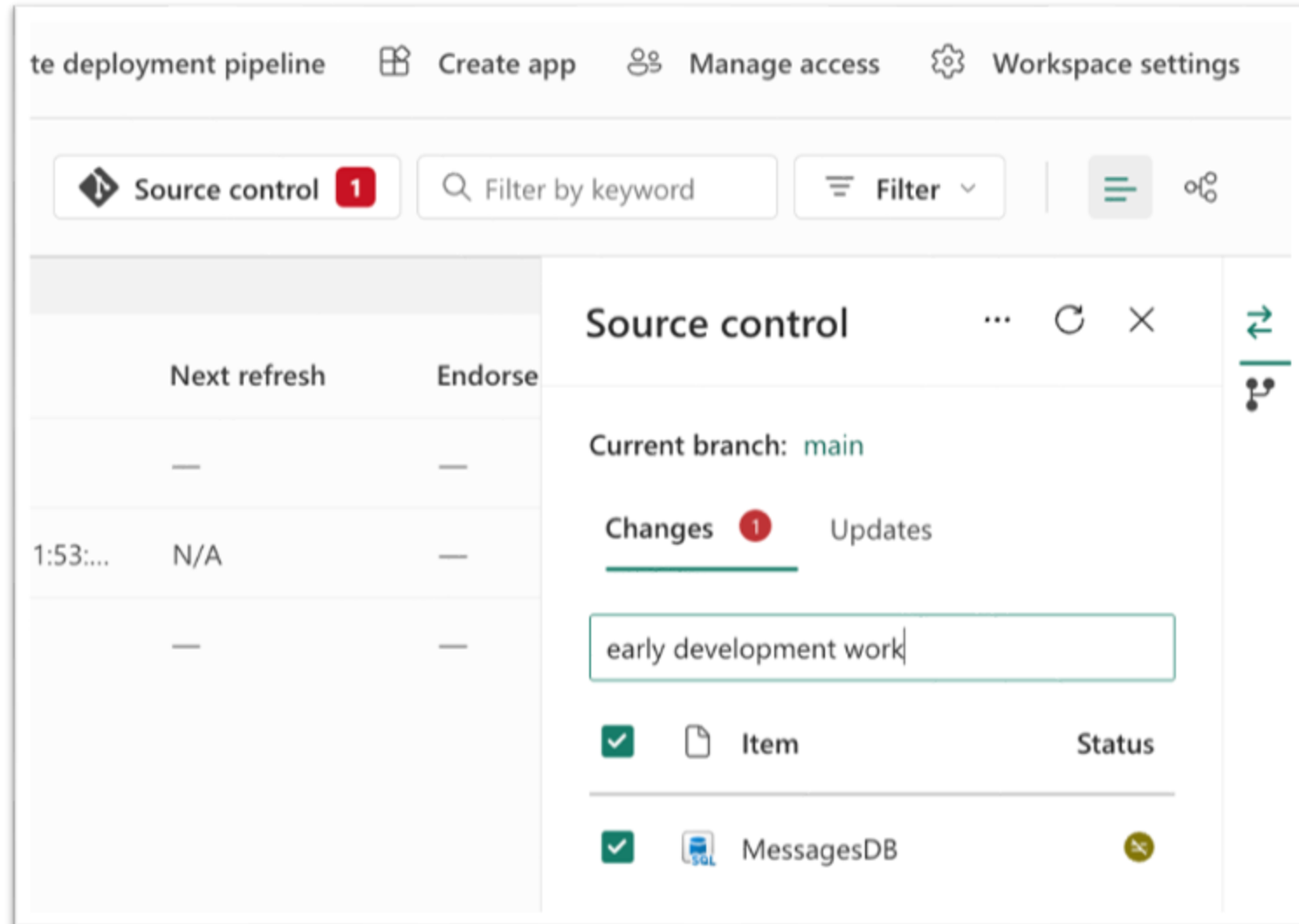
Fabric SQL DB CI/CD goals

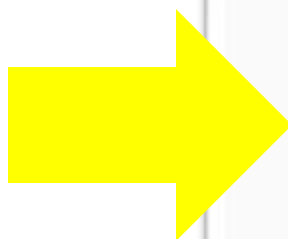
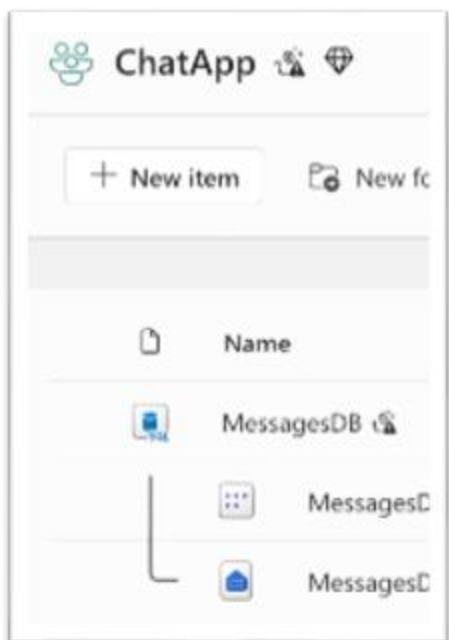


1. Get every database under source control
2. Make development databases self-service and disposable
3. Offer a low-friction mechanism to perform deployments and code quality checks without learning a new tool
4. Keep integration points open to more advanced CI/CD tools and workflows

Fabric - commit to source control

- Develop directly in the database
- Quickly iterate in early development
- Keep track of changes (commit along the way)





t App Demo / Repos / Files / Chat App Demo

Chat App Demo

- DatastoreWorkspace
 - MessagesDB.SQLDatabase
 - chatter
 - StoredProcedures
 - checkUserToken.sql
 - GetCurrentlyActivePlaces.sql
 - getLatestMessages.sql
 - GetUserInfo.sql
 - joinPlace.sql
 - newMessage.sql
 - newUser.sql
 - RevokeMessage.sql
 - updateUser.sql
 - Tables
 - Messages.sql**
 - PlaceMembership.sql
 - Places.sql
 - TextAnalysis.sql
 - Users.sql
 - Security
 - .platform
 - MessagesDB.sqlproj

Messages.sql

Contents History Compare Blame

```
1 CREATE TABLE [chatter].[Messages] (  
2   [MessageId]      INT          IDENTITY (1, 1) NOT NULL,  
3   [UserId]         UNIQUEIDENTIFIER NOT NULL,  
4   [PlaceId]        INT          NOT NULL,  
5   [CreatedDateTime] DATETIME2 (7) NOT NULL,  
6   [IsActive]       BIT          NOT NULL,  
7   [MessageContents] NVARCHAR (MAX) NOT NULL,  
8   PRIMARY KEY CLUSTERED ([MessageId] ASC),  
9   CONSTRAINT [FK_Messages_PlaceId] FOREIGN KEY ([PlaceId]) REFERENCES [chatter].[Places] ([PlaceId]),  
10  CONSTRAINT [FK_Messages_UserId] FOREIGN KEY ([UserId]) REFERENCES [chatter].[Users] ([UserId])  
11 );  
12  
13  
14 GO  
15  
16 CREATE NONCLUSTERED INDEX [IX_Messages_PlaceId_IsActive_CreatedDateTime]  
17   ON [chatter].[Messages]([PlaceId] ASC, [IsActive] ASC, [CreatedDateTime] ASC);  
18  
19  
20 GO  
21  
22
```

DatastoreWorkspace

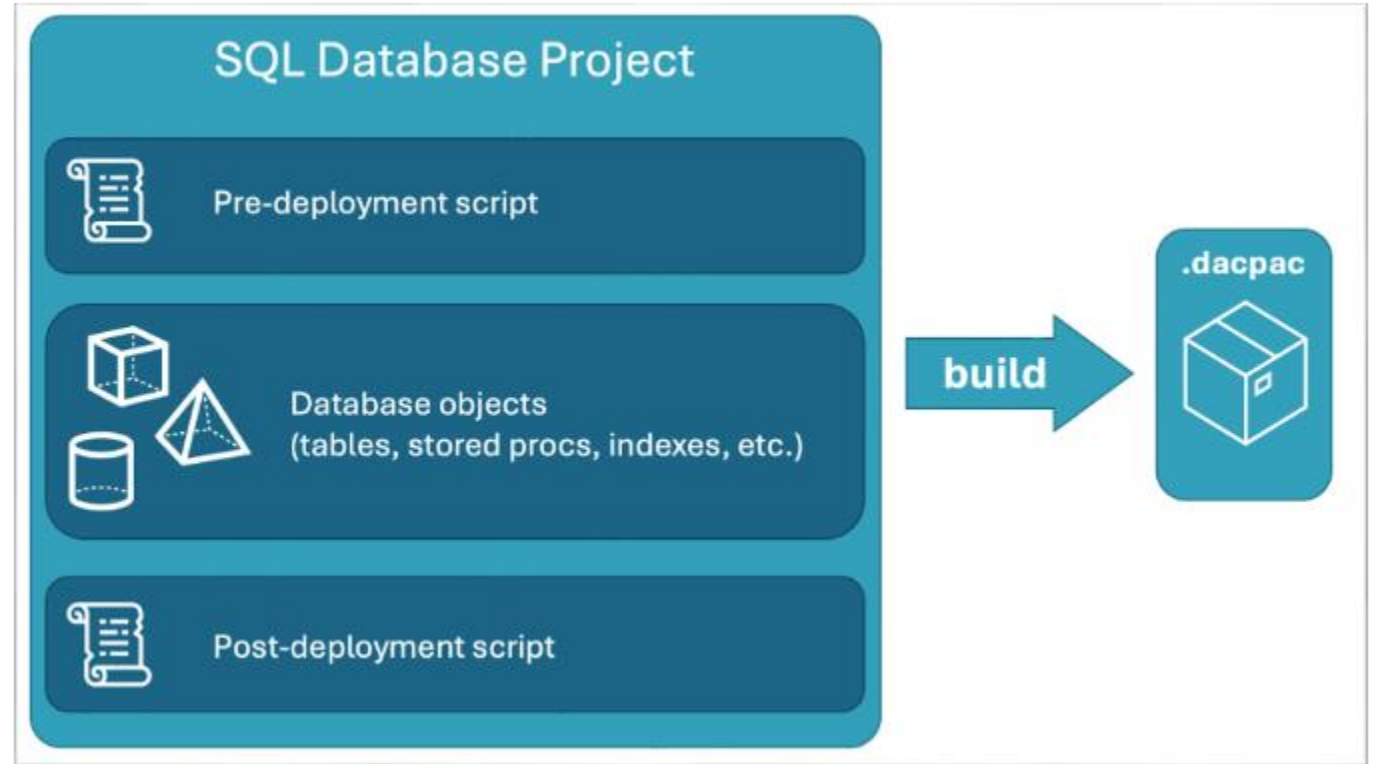
Contents History

Graph Commit

- early development work
78615567 Drew Skwiers-Koballa Today at 12:40 PM
- Committing 1 item from workspace 164f249e-b429-4351-8fd5-3c83ef37ad76
9bb0e9ba Drew Skwiers-Koballa Today at 12:14 PM
- Creating directory DatastoreWorkspace
42a5ede8 Drew Skwiers-Koballa Today at 12:14 PM

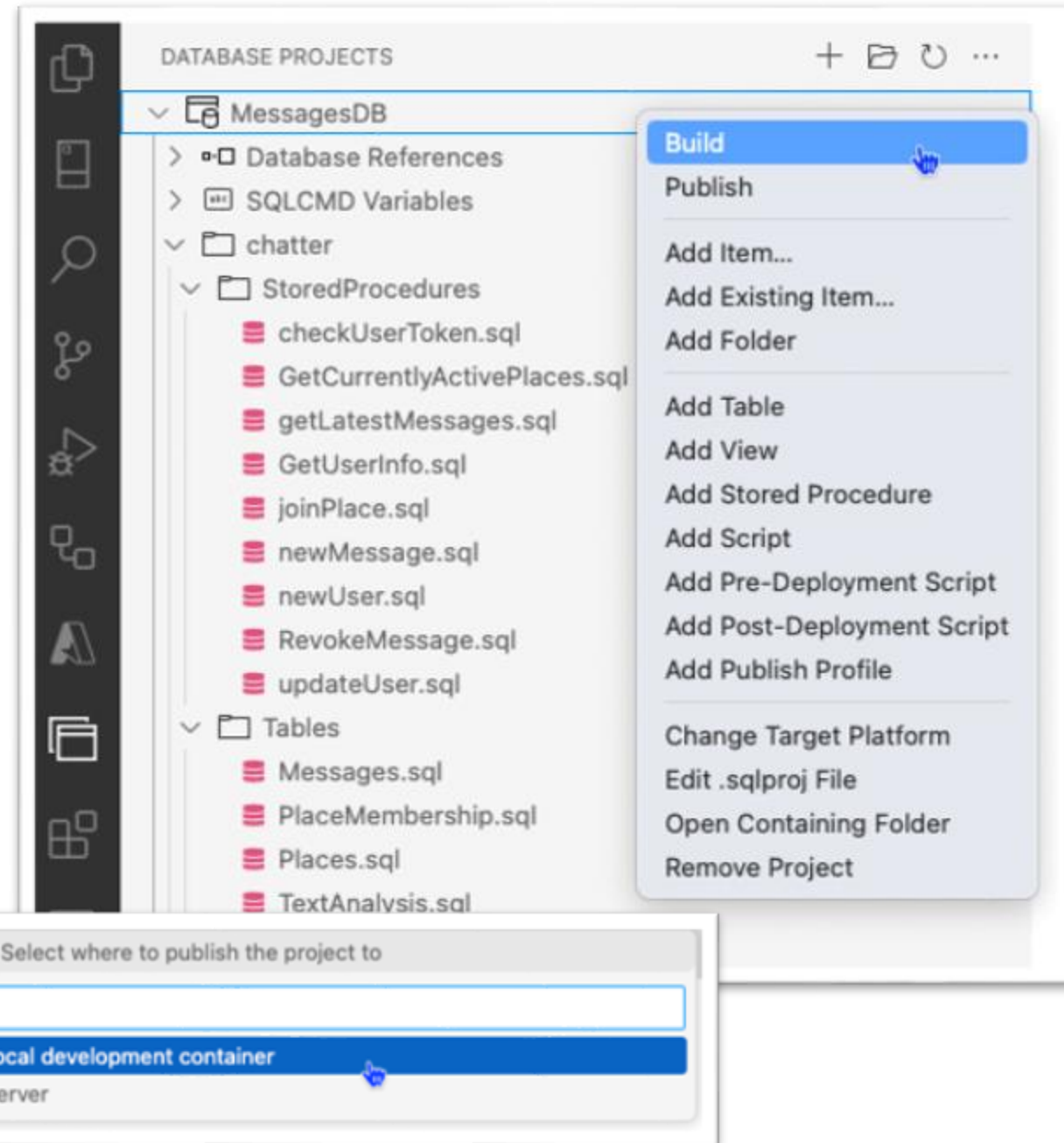
What's in the database code?

- SQL database project
- 1 file for each object
- Definitions as CREATE T-SQL statements
- Build process validates database model
- Portable and reusable artifact of database state



Develop in client tools

- Create new branch in source control
- Clone (copy) code to local machine
- Edit and build SQL database project in VS Code with mssql and SQL Database Projects extension
- Connect to Fabric or use a local instance for testing



Source control setup

1. Primary Fabric workspace where proof of concept (POC) work is taking place
2. Create an empty git repository (Azure DevOps or GitHub)
3. Connect workspace to main* branch of repo

**main or alternative branch based on your team's branching strategy*

Git integration (Preview)

Connect to Git to manage your code and back up your work. [Learn more](#)

 Connect Git provider and account

Provider
Azure DevOps

AAD account
drskwier@microsoft.com

Log out

Manage all accounts

 Connect Git repository and branch

Organization *

drskwier

Project *

Chat App Demo

Git repository * ⓘ

Chat App Demo

Branch * ⓘ

main

Git folder ⓘ

Enter name of folder

Connect and sync

Cancel

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Drew Sessions ...

Sessions_data base

...

Analyze

Microsoft

Analyze

Drew Sessions Finder

Search

Trial: 20 days left

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Create deployment pipeline

Create app

Manage access

Workspace settings

+ New item

New folder

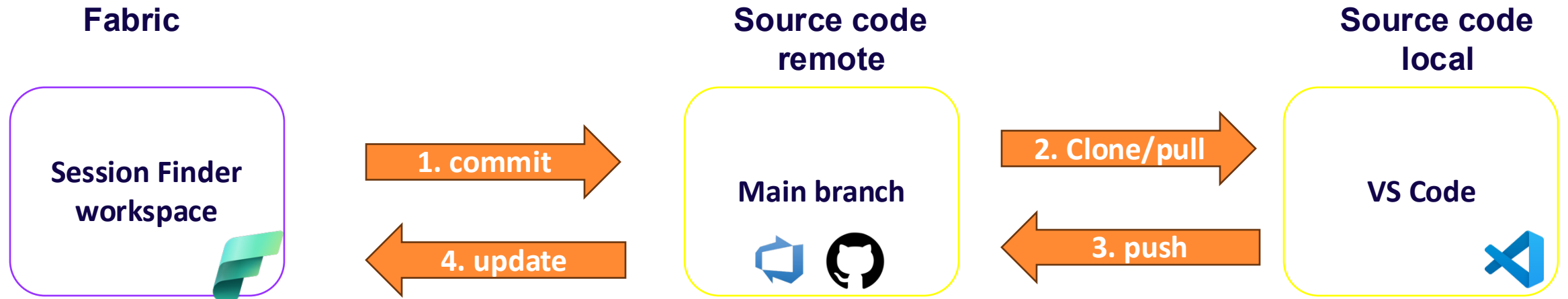
Upload

Filter by keyword

Filter

	Name	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
	Dataflow 1	Dataflow Gen2	—	Drew Skwiers...	11/1/2024, 9:52...	N/A	—	Confidential\...	ⓘ
	find_sessions_wrapper	API for Graph...	—	Drew Skwiers...	—	—	—	Confidential\...	ⓘ
	Sessions Finder Notebook	Notebook	—	Drew Skwiers...	—	—	—	Confidential\...	ⓘ
	Sessions_database	SQL Database	—	Drew Skwiers...	—	—	—	Confidential\...	ⓘ
	Sessions_database	Semantic mo...	—	Drew Session...	11/1/2024, 9:19...	N/A	—	Confidential\...	ⓘ
	Sessions_database	SQL analytics...	—	Drew Skwiers...	—	—	—	Confidential\...	ⓘ

Source control integration in action



Microsoft

Analyze

Drew Sessions Finder

Search

Trial: 20 days left

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Drew Sessions Finder

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+ New item

New folder

Upload

Source control 0

Filter by keyword

Filter

	Name	Git status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
	Dataflow 1	Unsupported	Dataflow G...	—	Drew Skwie...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	find_sessions_wrapper	✓ Synced	API for Gra...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions Finder Notebook	✓ Synced	Notebook	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	✓ Synced	SQL Databa...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	—	Semantic m...	—	Drew Sessio...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	Sessions_database	—	SQL analyti...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ

dev/sessions-finder

Last synced: 11/1/2024 at 11:01 AM

e72bf746

Microsoft

Analyze

Drew Sessions Finder

Search

Trial: 20 days left

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Source control 0

Filter by keyword

Filter

	Name	Git status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
	Dataflow 1	Unsupported	Dataflow G...	—	Drew Skwie...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	find_sessions_wrapper	✓ Synced	API for Gra...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions Finder Notebook	✓ Synced	Notebook	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	✓ Synced	SQL Databa...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	—	Semantic m...	—	Drew Sessio...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	Sessions_database	—	SQL analyti...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ

dev/sessions-finder

Last synced: 11/1/2024 at 11:04 AM

37b14bba

msdata / Database Systems / Repos / Files / SQLDB Native Fabric ALM Testing

Database Systems

Overview

Boards

Repos

Files

Commits

Pushes

Branches

Tags

Pull requests

EzCab

EzCab-Test

Ownership

Pipelines

Test Plans

Artifacts

Compliance

Project settings

SQLDB Native Fabric ALM Testing

FabricWorkspace

DataflowsStagingLakehouse.Lakehou

DataflowsStagingWarehouse.Warehous

find_sessions_wrapper.GraphQLApi

Sessions Finder Notebook.Notebook

Sessions_database.SQLDatabase

dbo

Sequences

StoredProcedures

Tables

searched_text.sql

sessions_feedback.sql

sessions_speakers.sql

sessions.sql

speaker_profiles.sql

speakers.sql

Security

.gitignore

.platform

Sessions_database.sqlproj

Readme.md

dev/sessions-finder

Sessions_database.SQLD...

dbo

Tables

sessions_feedback.sql

sessions_feedback.sql

Contents

History

Compare

Blame

You updated dev/sessions-finder 3m ago

Create a pull request

Full history (simplify merges)

Author

Graph	Commit	Change	Pull Request	Status
	improve feedback - datetime and... a359926e Drew Skwiers-Koballa Today at 11:22 AM	edit		
	adding session feedback 37b14bba Drew Skwiers-Koballa Today at 11:04 AM	add		

Microsoft

Analyze

Drew Sessions Finder

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Trial: 20 days left

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Source control 1

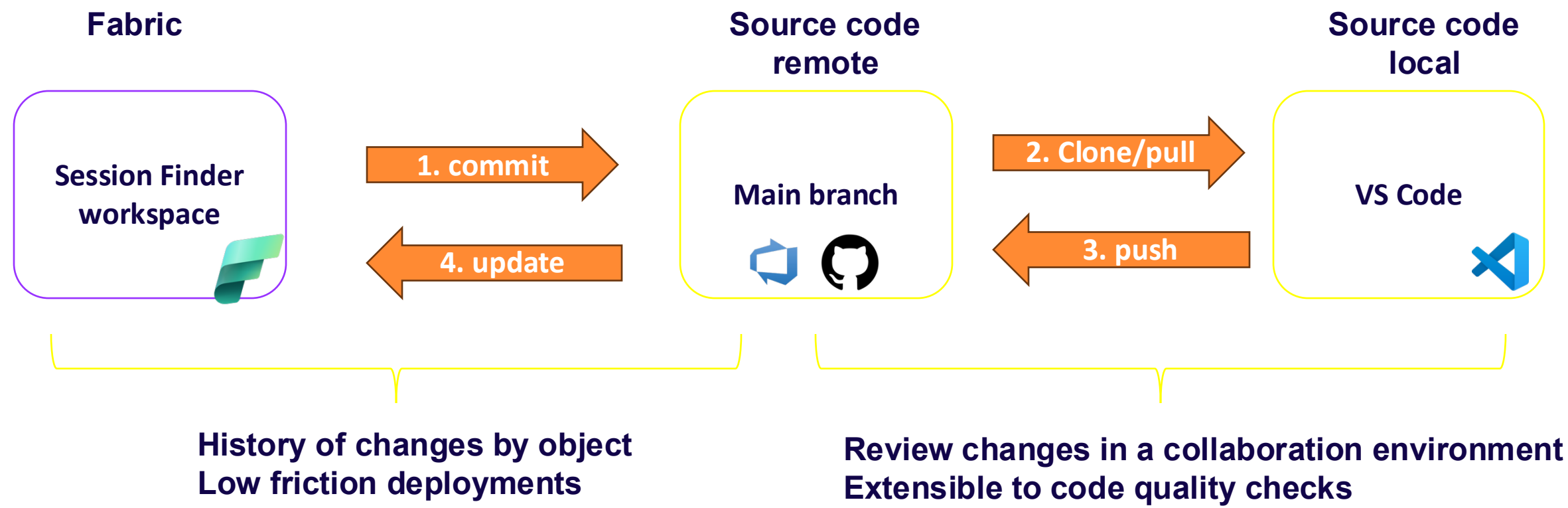
Filter by keyword

Filter

	Name	Git status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
	Dataflow 1	Unsupported	Dataflow G...	—	Drew Skwie...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	find_sessions_wrapper	✓ Synced	API for Gra...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions Finder Notebook	✓ Synced	Notebook	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	Uncommitted	SQL Databa...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	—	Semantic m...	—	Drew Sessio...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	Sessions_database	—	SQL analyti...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ

dev/sessions-finder Last synced: 11/1/2024 at 11:27 AM a359926e

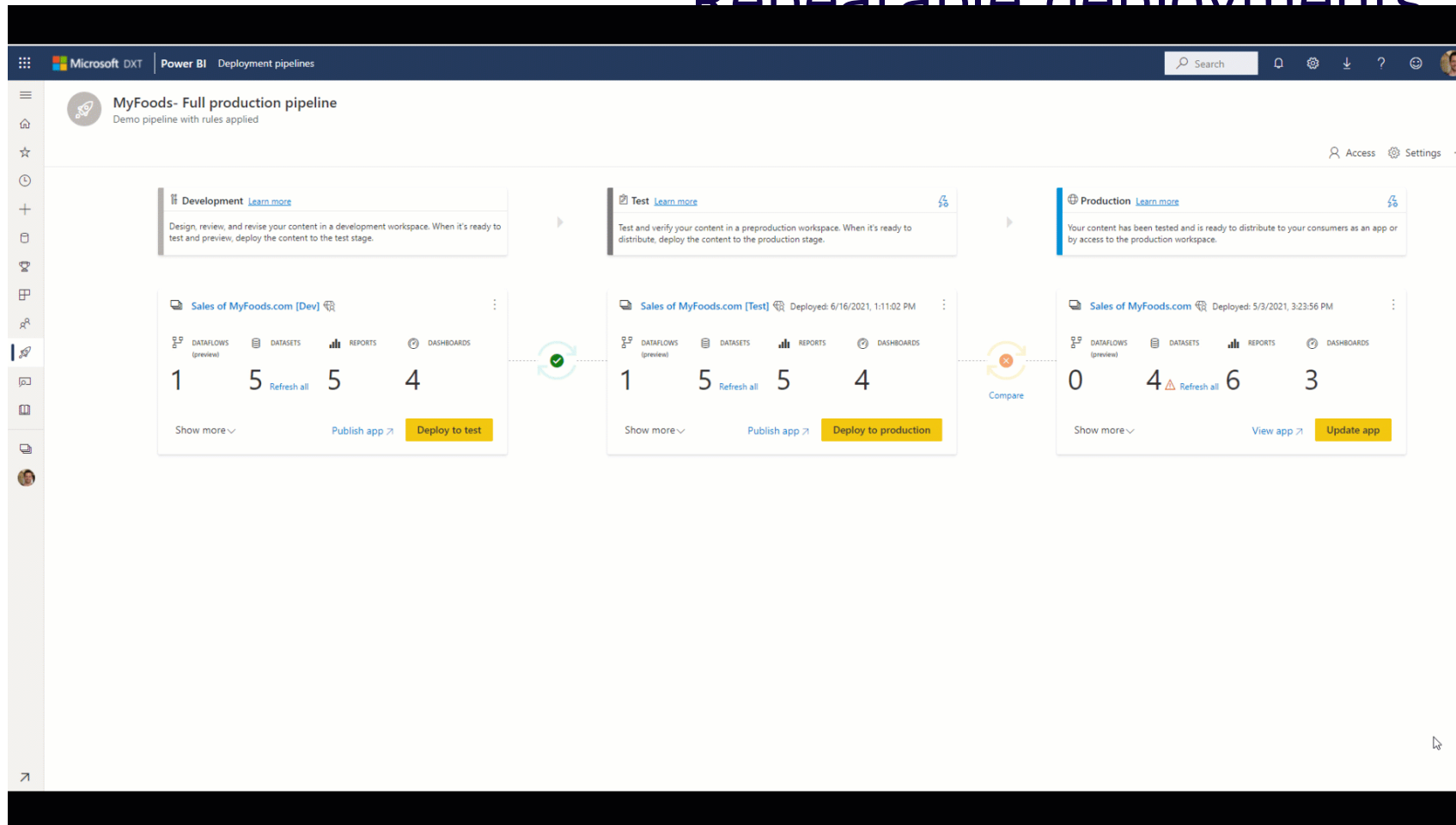
Benefits from source control integration



Fabric Deployment pipelines

- Multi-workspace deployment process
- Dev/Test/Prod environment automation

Repeatable deployments



Setup Azure DevOps Identity



Azure
DevOps
Pipelines



Azure

New service connection

Choose a service or connection type

Search connection types

- ☐ Azure Classic
- ☐ Azure Repos/Team Foundation Server
- ☒ Azure Resource Manager
- ☐ Azure Service Bus
- ☐ Bitbucket Cloud

New Azure service connection

Azure Resource Manager

Authentication method

- ☒ Workload Identity federation (automatic) Recommended
- ☐ Workload Identity federation (manual)
- ☐ Service principal (automatic)
- ☐ Service principal (manual)
- ☐ Managed identity
- ☐ Publish Profile

[Need help choosing a connection type?](#) Back Next

New Azure service connection

Azure Resource Manager using Workload Identity federation with OpenID Connect (automatic)

Scope level

- ☒ Subscription
- ☐ Management Group
- ☐ Machine Learning Workspace

Subscription

Resource group

drskwier-chat-app-hackathon

Details

Service connection name

devops-demo

Description (optional)

[Home](#) >

drskwier-Chat App Demo-d

Search

Delete Endpoints Preview features

Overview

Quickstart

Integration assistant

Diagnose and solve problems

> Manage

> Support + Troubleshooting

^ **Essentials**

Display name : **drskwier-Chat App Demo-d51**

Application (client) ID :

Object ID :

Directory (tenant) ID :

Supported account types : [My organization only](#)

Setup Azure DevOps pipeline



🔒 Manage SQL security



Manage access

Add or remove users for this role.

Add people, groups or apps

drskwier-Chat App Demo-d51



drskwier-Chat App Demo-d51

Add

Clear



```
40  # this connection settings would usually be in variables
41  Settings
42  - task: SqlAzureDacpacDeployment@1
43    inputs:
44      azureSubscription: 'devops-demo'
45      AuthenticationType: 'aadAuthenticationIntegrated'
46      ServerName: 'fleet-srv-767c0a4b-beec-4d0a-9c10-333efe44de71-a08b9015aef7.database.windows.net'
47      DatabaseName: 'MessagesDB-91784e0c-92f1-4c30-b3c7-2e40acd5be9c'
48      deployType: 'DacpacTask'
49      DeploymentAction: 'Publish'
50      DacpacFile: '$(Pipeline.Workspace)/s/DatastoreWorkspace/MessagesDB.SQLDatabase/bin/Debug/MessagesDB.dacpac'
51      IpDetectionMethod: 'AutoDetect'
```



SQL database in Fabric CI/CD ecosystem

Development lifecycle process

Iterative development Validation & deployment

Tooling

In Fabric

Workspace check into
and update from
source control

Multi-workspace
deployment pipelines

Outside Fabric

SQL projects in VS
Code and Visual
Studio

SqlPackage publish,
Azure DevOps
pipelines



Fabric deployment pipelines



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Source control 0

Filter by keyword

Filter

	Name	Git status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
	Dataflow 1	Unsupported	Dataflow G...	—	Drew Skwie...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	find_sessions_wrapper	✓ Synced	API for Gra...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions Finder Notebook	✓ Synced	Notebook	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	✓ Synced	SQL Databa...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	—	Semantic m...	—	Drew Sessio...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	Sessions_database	—	SQL analyti...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ

dev/sessions-finder

Last synced: 11/1/2024 at 11:56 AM

6e98d029

Microsoft

Sessions_database

Confidential\Microsoft Extended

Search

Trial: 20 days left

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Replication

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Get data

New Query

Templates

Open in

New API for GraphQL

Performance summary

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Sessions_database

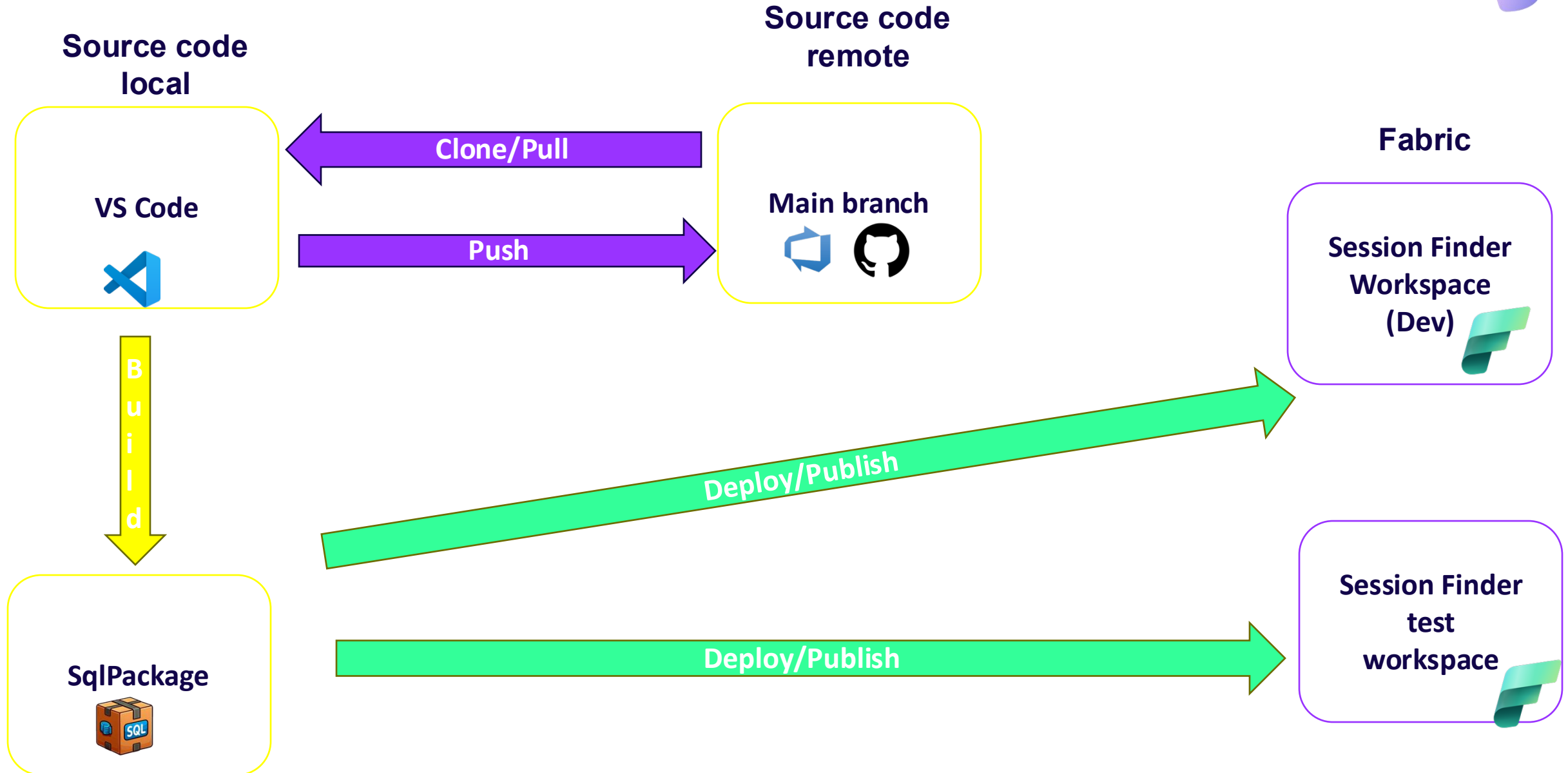
Queries

Query, preview, or connect your data

Interact with your data by starting a query, creating database objects with a template, or connecting the database to your app with a GraphQL API. You can also preview your data by opening a table or view.

+ New

SqlPackage deployments



SOURCE CONTROL

SOURCE CONTROL

Message (⌘Enter to commit ...)

✓ Commit

Changes

DataflowsStagingWarehouse.s...

SOURCE CONTROL ... Auto

improve fe... @ dev/sessions-fi...

- adding session feedback Drew Skw...
- dataflow staging warehouse Drew ...
- Committing 5 items from workspac...
- Creating directory FabricWorkspac...
- Updated owners.txt Charles Gagnon
- Updated README.md Charles Gagn...
- Adding owners.txt containing Servi...
- Adding scaffolding to git repo Char...

sessions_feedback.sql

FabricWorkspace > Sessions_database.SQLDatabase > dbo > Tables > sessions_feedback.sql

```
1 CREATE TABLE [dbo].[sessions_feedback] (
2     [id] INT DEFAULT (NEXT VALUE FOR [global_id]) NOT NULL,
3     [session_id] INT NOT NULL,
4     [session_score] INT NOT NULL,
5     [speaker_score] INT NOT NULL,
6     [comment] NVARCHAR (MAX) NULL,
7     [review_time] DATETIME2 NOT NULL,
8
9     PRIMARY KEY CLUSTERED ([id] ASC),
10    FOREIGN KEY ([session_id]) REFERENCES [sessions] ([id])
11 );
12
13
14 GO
15
16
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS ...

Database Projects

```
TutorialWork/Workshop/FabricWorkspace/Sessions_database.SQLDatabase/Sessions_database.
sqlproj]
```

```
stdout: /Users/drewsk/Documents/CodeRepos/TutorialWork/Workshop/FabricWorkspace/
```

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Source control 0

Filter by keyword

Filter

	Name	Git status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
	Dataflow 1	Unsupported	Dataflow G...	—	Drew Skwie...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	find_sessions_wrapper	✓ Synced	API for Gra...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions Finder Notebook	✓ Synced	Notebook	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	✓ Synced	SQL Databa...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ
	Sessions_database	—	Semantic m...	—	Drew Sessio...	11/1/2024, 9:...	N/A	—	Confidentia...	ⓘ
	Sessions_database	—	SQL analyti...	—	Drew Skwie...	—	—	—	Confidentia...	ⓘ

dev/sessions-finder

Last synced: 11/1/2024 at 11:56 AM

6e98d029

Lab Two

- Test the deployment with a SQL Server 2022 container in a pipeline
-

Advanced Topics

Watch out for these SQL projects pitfalls!

Managing users/roles

- Focus on managing DB roles
- Consider segmenting user assignment to roles in another cadence
- Change controls are ideally integrated into user assignment process

Schema compare gaps

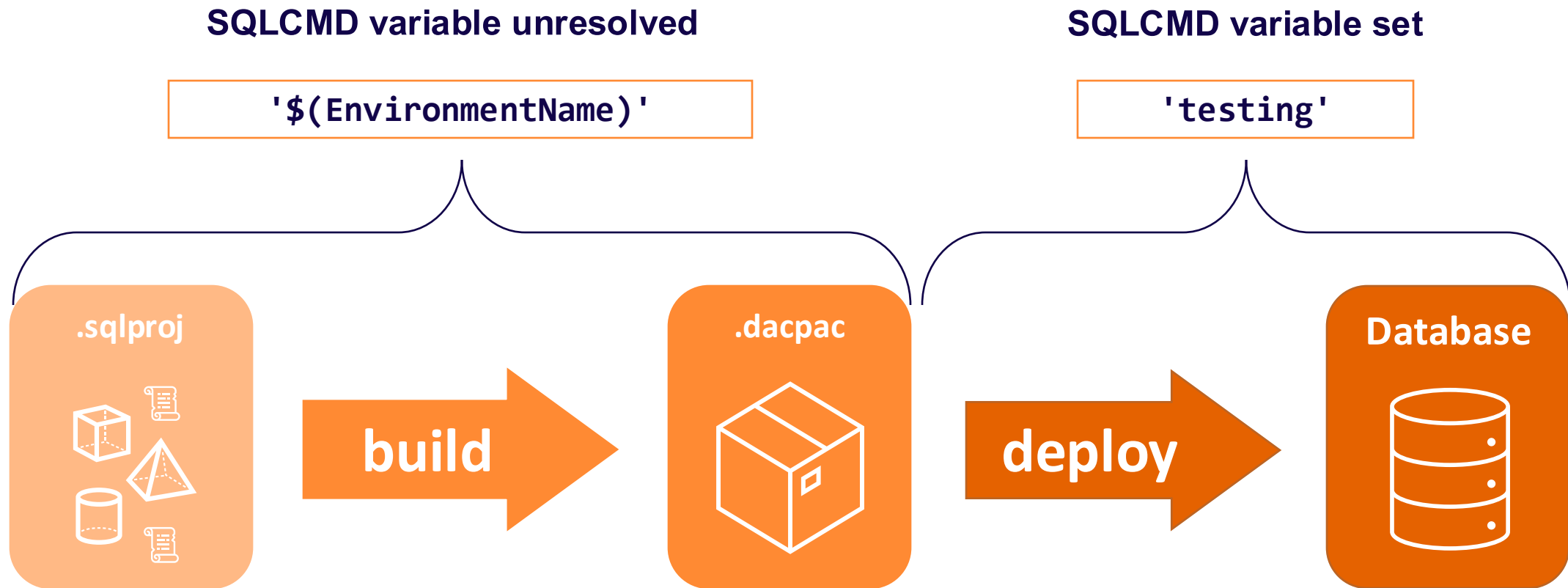
- SQL applies a default name to unnamed constraints
- Schema compare will display the object as having a difference when dealing with unnamed constraints and SQLCMD variables

Connectivity to DB

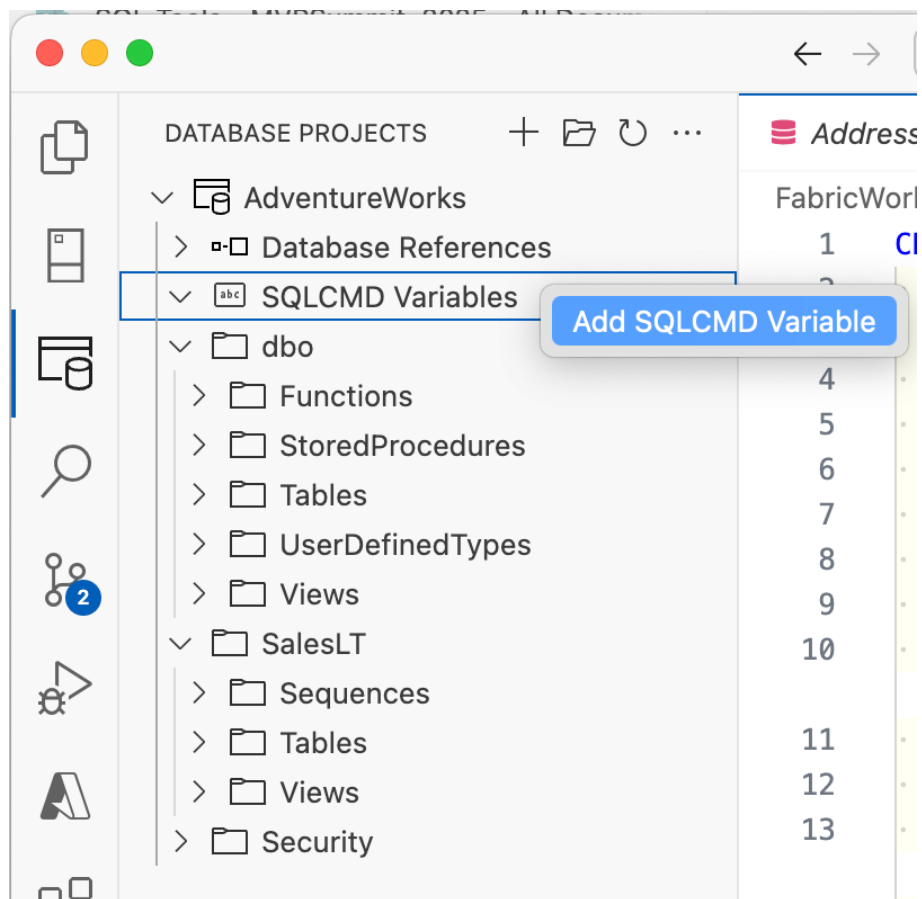
- Azure SQL DB open to all Azure services is not recommended except for early dev
- SQL Server instances usually have full network isolation
- Consider **self-hosted runners** for controlled environments

SQLCMD variables

- Placeholder in any SQL projects T-SQL code that is only applied during deployment
- Control IF/ELSE statements, fill in default values, set database name in 3-part naming



SQLCMD variables



.sqlproj file

```
<ItemGroup>
  <SqlCmdVariable
    Include="StudentPassword">
    <Value>$(SqlCmdVar__1)</Value>
    <DefaultValue>password</DefaultValue>
  </SqlCmdVariable>
</ItemGroup>
```

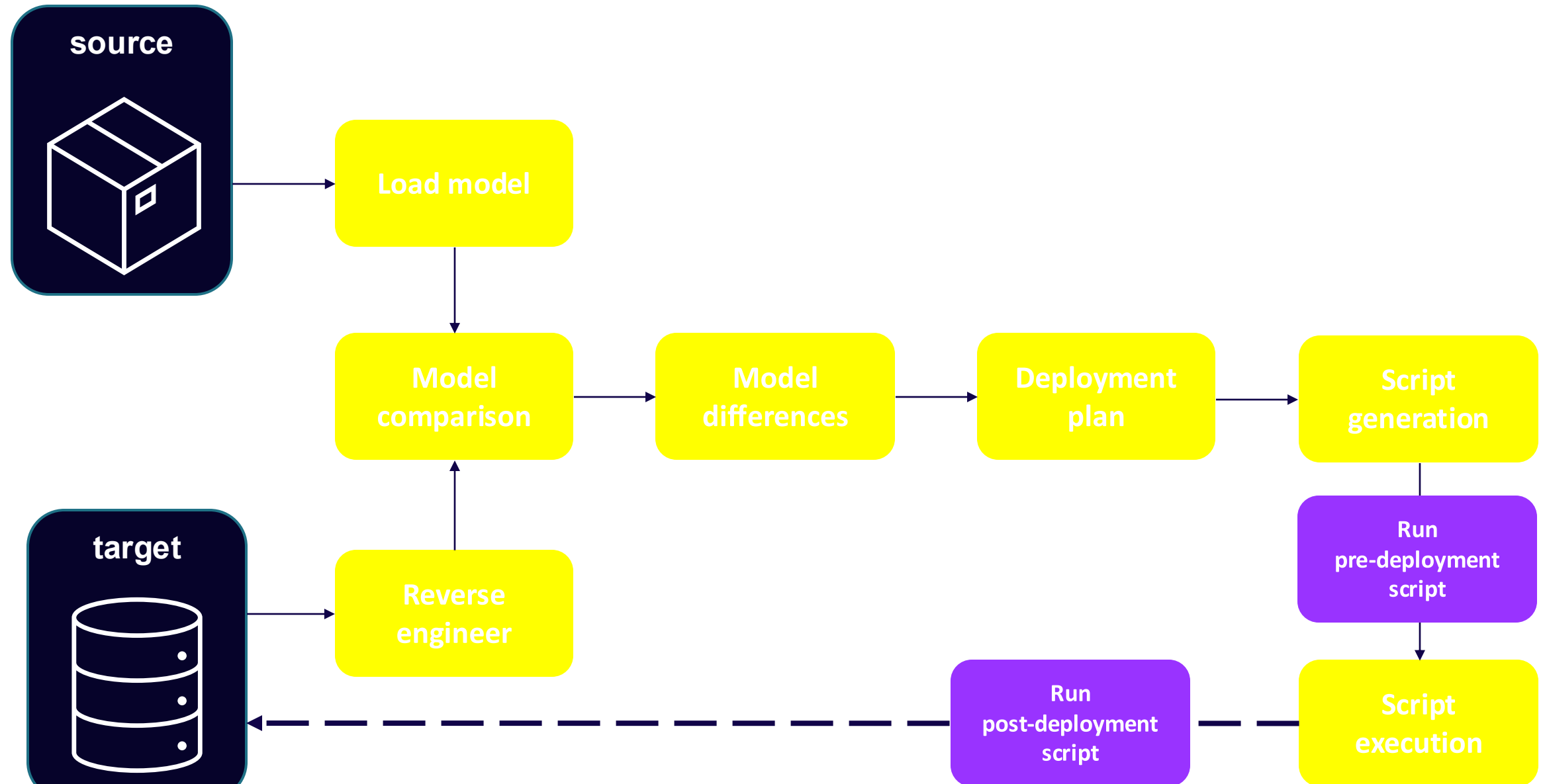
Use in scripts

```
CREATE USER [student] WITH
PASSWORD = '$(StudentPassword)';
```

Pre- and post-deployment scripts

- Non-object SQL files in a SQL project that are run with each deployment
- First line of flexibility with SQL projects to support more situations
- Partially enabled for Fabric source control integration (run once, will be removed on next commit from Fabric)

Dacpac deployment behind the scenes

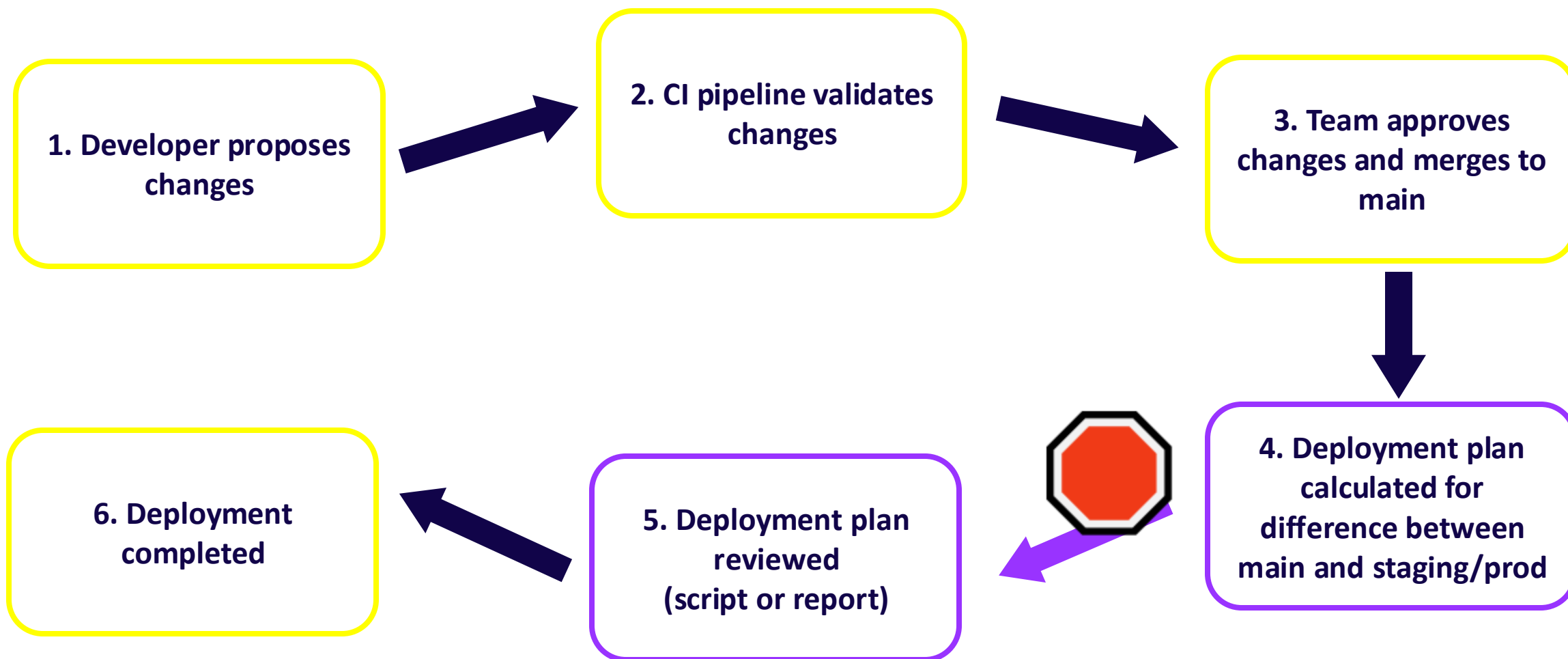


Pre/post-deployment scripts in .sqlproj

```
<?xml version="1.0" encoding="utf-8" ?>
<Project DefaultTargets="Build" >
  <Sdk Name="Microsoft.Build.Sql" Version="1.0.0" />
  <PropertyGroup>
    <Name>Wingtips</Name>
    <DSP>Microsoft.Data.Tools.Schema.Sql.SqlAzureV12DatabaseSchemaProvider</DSP>
    <ModelCollation>1033,CI</ModelCollation>
    <ProjectGuid>{00000000-0000-0000-0000-000000000000}</ProjectGuid>
  </PropertyGroup>
  <ItemGroup>
    <PreDeploy Include="MyPredeployScript.Sql" />
  </ItemGroup>
</Project>
```


Static data management

- (other than pre/post-deployment scripts...)
- BCP
- External tables
- Manual notebook execution
- Custom process with backup restore and data selection



100 Gate pipelines

- Automated does not have to be automatic
- Use **GitHub environments** or **Azure DevOps approvals** to gate a workflow

← Deploy SQL project
🕒 **Deploy SQL project #4**

Summary

Jobs

- ✅ build
- 🕒 deploy

Run details

- 🕒 Usage
- 📄 Workflow file

Manually triggered 4 minutes ago

👤 dzsquared → a998f40 **main**

Status **Waiting**

Total duration **—**

Artifacts **1**

🔗 dzsquared requested your review to deploy to **ProductionDeployment**

publish.yml
on: workflow_dispatch

✅ build 3m 34s → 🕒 deploy
ProductionDeployment waiting for review

Deployment protection rules
Reviewers, timers, and other rules protecting deployments in this run

Event Environments

Summary Scans Code Coverage

Repository and version

📦 Library

🌐 main 🔗 9631475c

Time started

🕒 Just now

🕒 3 of 10 checks need your review before this run can continue

Stages Jobs

🕒 **Deploy_Staging**

Waiting

📄 2 checks in progress

Checks for Deploy_WUS2
Timeout in 23h

- 🕒 **Approval** 🧑 Environment [FabrikamFiber Feed WUS1](#)
Waiting for approval • "Approve after other checks passed"
- 🕒 **Approval** 🧑 Variable group [AzureFunctionKeys](#)
Waiting for approval
- 🕒 **Business Hours Ren...** 🧑 Variable gr... [AzureFunction...](#)
In progress
- ✅ **Branch control** 🧑 Environment [FabrikamFiber Feed WUS1](#)
Passed

Comment (optional) Reject **Approve**

Network access challenges

Any Azure allowed

- Azure SQL Database, SQL database in Fabric
- Effectively the same as public access
- Caution
- Early development loops

Firewall rules

- Add/remove just-in-time (JIT)
- Leverages EntraID access to Azure resource management

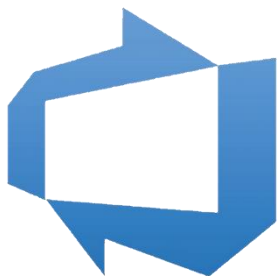
Private endpoints

- Isolated networks
- No incoming traffic allowed
- Deploy a pipeline runner in the network

"Easy mode" (for development)

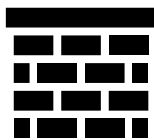


GitHub actions



Azure DevOps
pipelines

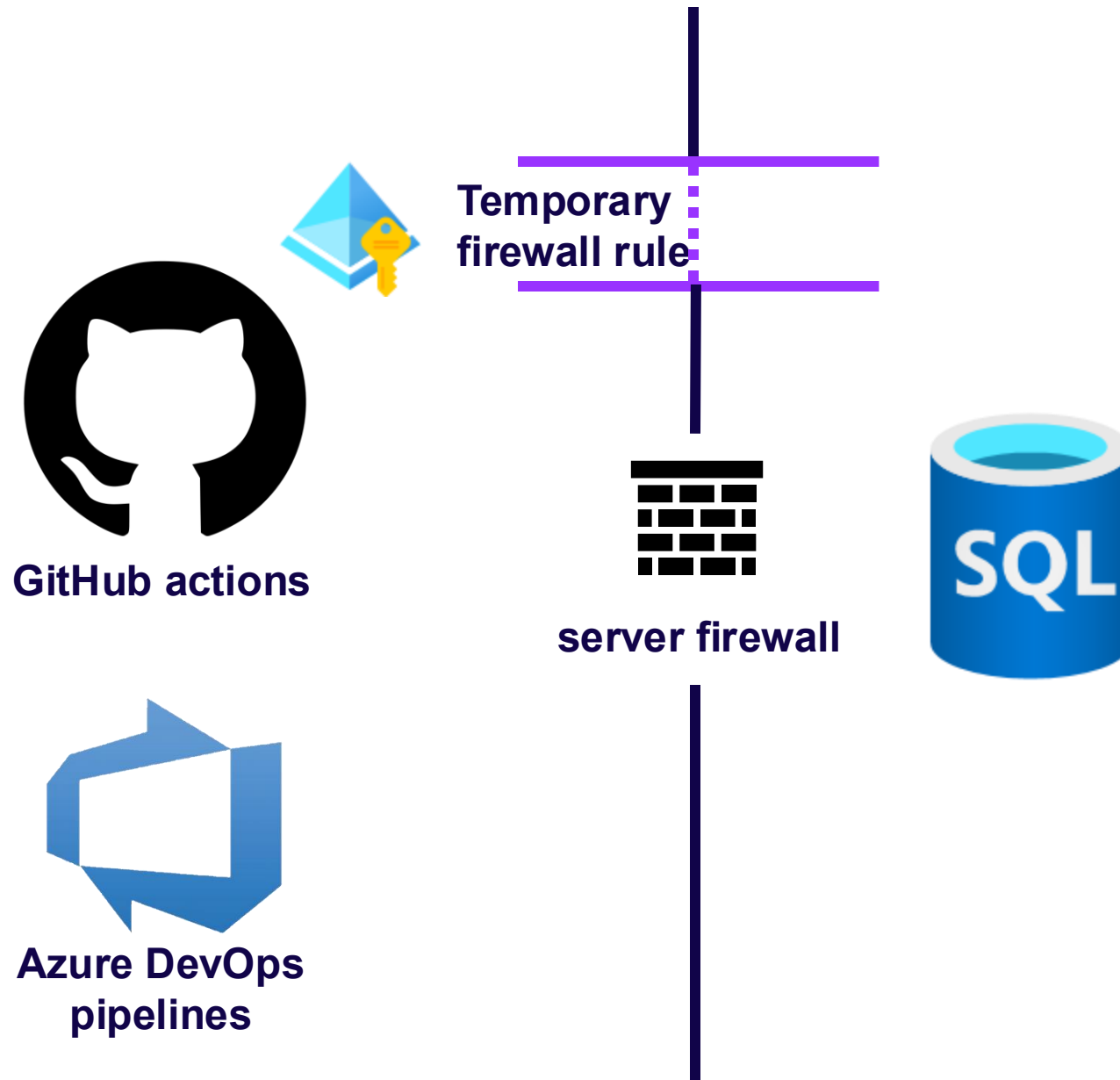
Allow Azure services
access



server firewall

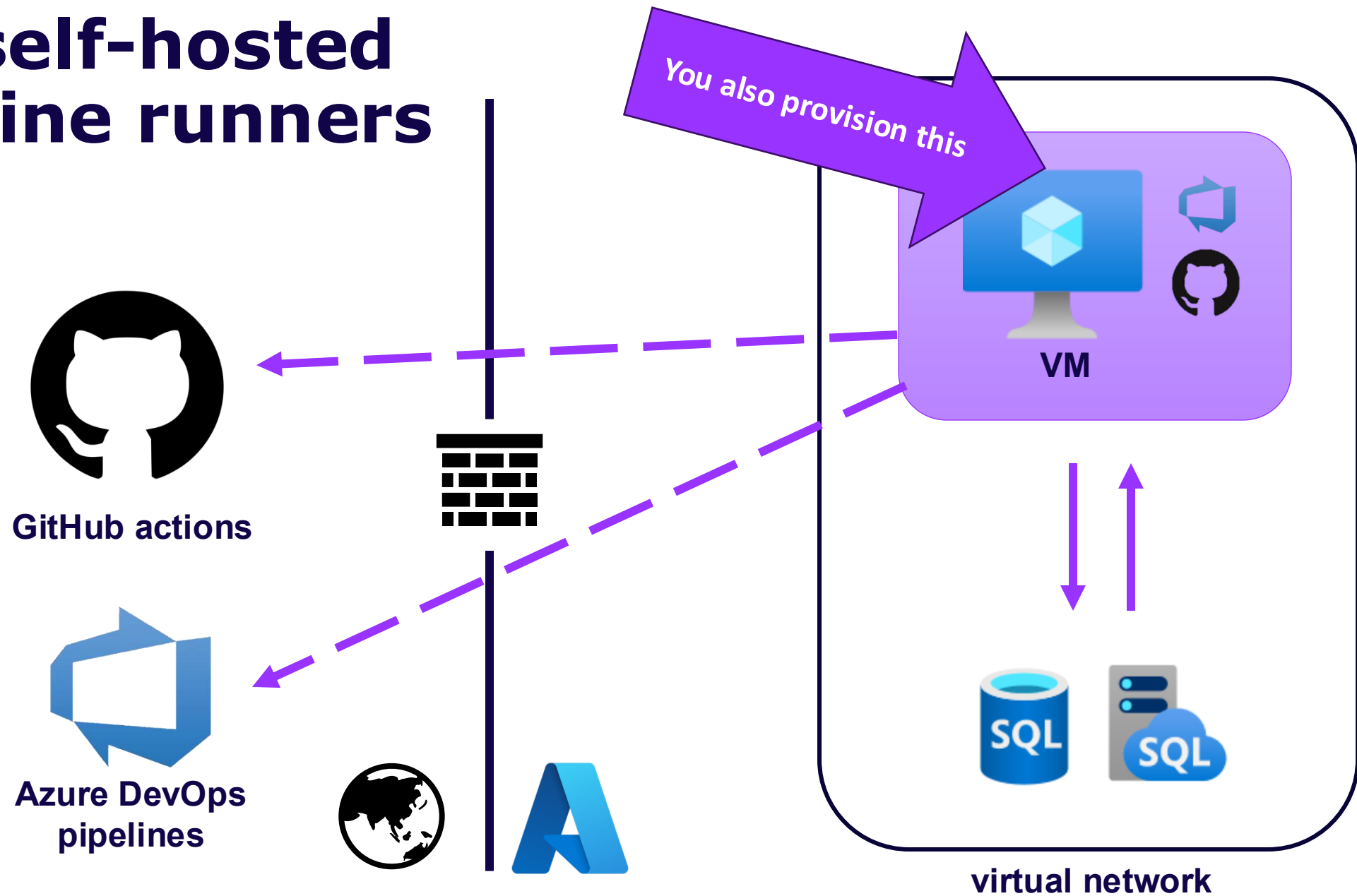


- GitHub and Azure DevOps pipelines are hosted in Azure infra
- "Allow Azure services access" permits any pipeline to reach your database
- Think about the consequences of that access



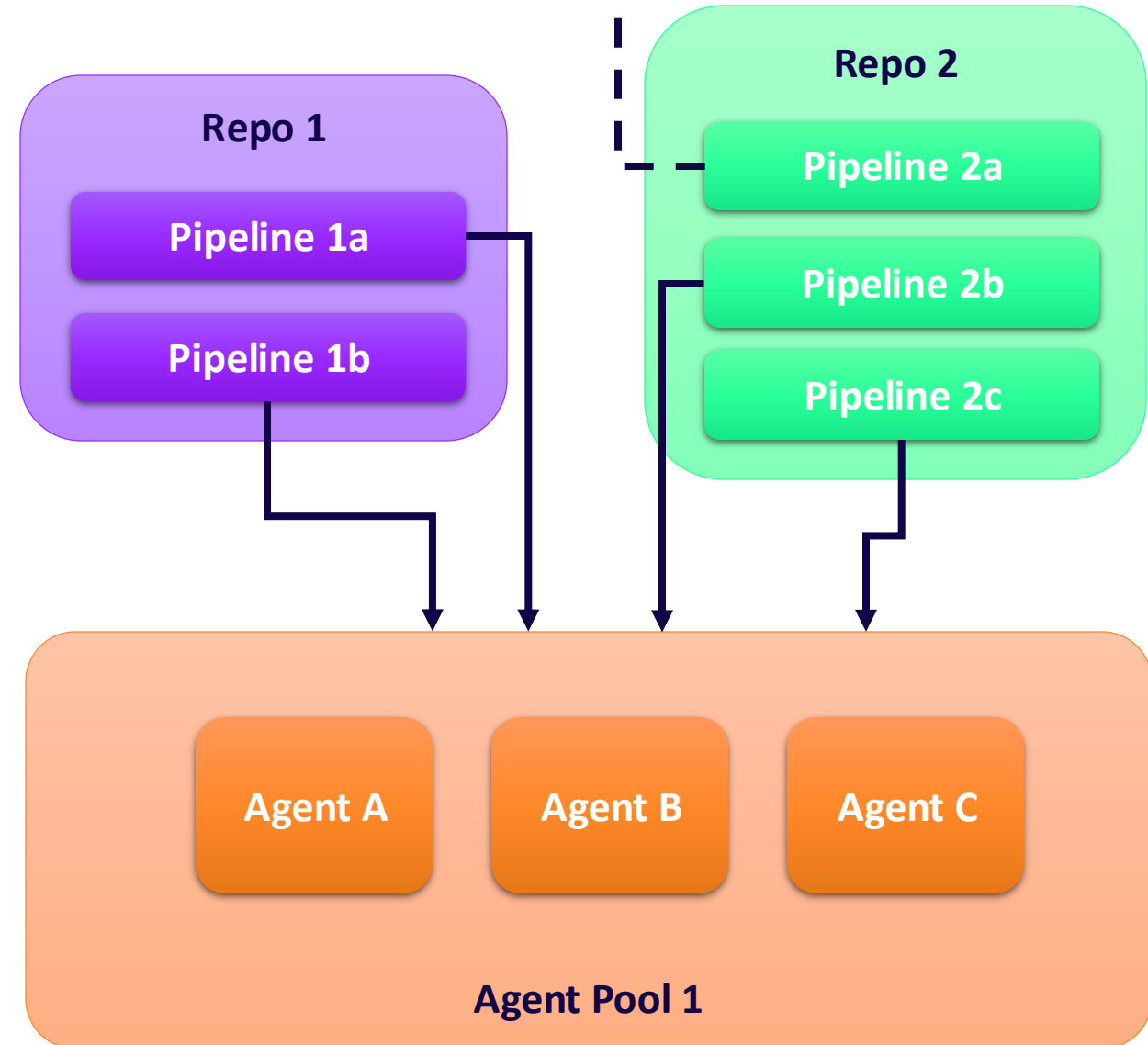
- Leverage managed identities as a best practice
- Add+remove a temporary firewall rule
- Tasks in GitHub and ADO incorporate a firewall rule for Azure SQL DB
- Use az cli or Azure PowerShell to directly modify firewall rules
- Azure SQL MI public networking can be done but is much more complex

104 Network isolation use self-hosted pipeline runners



Self-hosted automation environments

- **Enables network isolation**
- Runner only needs to be able to call out to check for pending jobs
- Same runner/agent can be used across many pipelines
- Groups of runners/agents are pooled to share workloads
- You manage the OS, installed software, availability, etc
- Commonly configured as VMs or containers



Add self-hosted runner to a single GitHub repo

<> Code

Issues

Pull requests 1

Actions

Projects

Security

Insights

Settings

General

Access

Collaborators

Code and automation

Branches

Tags

Rules

Actions

General

Runners

Webhooks

Environments

Codespaces

Pages

Security

Advanced Security

Deploy keys

Secrets and variables

Runners / Add new self-hosted runner · dzsquared/wingtips-devops-samples

Adding a self-hosted runner requires that you download, configure, and execute the GitHub Actions Runner. If you do not already have an existing volume licensing agreement for your GitHub purchases, by downloading and configuring the GitHub Actions Runner, you agree to the [GitHub Customer Agreement](#).

Runner image

☐ macOS

☐ Linux

☒ Windows

Architecture

x64

Download

We recommend configuring the runner under "\actions-runner". This will help avoid issues related to service identity folder permissions and long path restrictions on Windows.

```
# Create a folder under the drive root
$ mkdir actions-runner; cd actions-runner

# Download the latest runner package
$ Invoke-WebRequest -Uri https://github.com/actions/runner/releases/download/v2.323.0/actions-runner-win-x64-2.323.0.zip -OutFile actions-runner-win-x64-2.323.0.zip

# Optional: Validate the hash
```

EXPLORER

...

OPEN EDITORS

! local-environment-update.yml .github/wo...

NewVenue.sql Wingtips/dbo/StoredProced...

Venues.sql Wingtips/dbo/Tables

X Dockerfile LocalRunner

WINGTIPS-DEVOPS-SAMPLES

> .github

> FabricWorkspace

LocalRunner

Dockerfile

\$ entrypoint.sh

> testWingtips

Wingtips

> bin

> control

dbo

StoredProcedures

DeleteEvent.sql

DeleteVenue.sql

NewVenue.sql

ResetEventDates.sql

Tables

Countries.sql

Customers.sql

Events.sql

EventSections.sql

Sections.sql

OUTLINE

TIMELINE

! local-environment-update.yml

NewVenue.sql

Venues.sql

Dockerfile X

LocalRunner > Dockerfile > ...

1 FROM ghcr.io/actions/actions-runner:2.323.0

2

3 USER root

4

5 # install dotnet sdk and sqlpackage

6 RUN apt-get update && apt-get install -y dotnet-sdk-8.0 && \

7 ... # apt-get install sqlcmd && \

8 ... dotnet tool install --tool-path /usr/local/bin/ Microsoft.SqlPackage

9

10 COPY entrypoint.sh ./entrypoint.sh

11 RUN chmod +x ./entrypoint.sh

12

13 USER runner

14

15 ENTRYPOINT ["/entrypoint.sh"]

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

AZURE

SPELL CHECKER

...

zsh - Wingtips + - [] [] ... X

tips.dll

Wingtips -> /Users/drewsk/Documents/CodeRepos/SampleSandbox/wingtips-devops-samples/Wingtips/bin/Debug/Wingtips.dacpac

Build succeeded.

0 Warning(s)

0 Error(s)

Time Elapsed 00:00:05.59

→ Wingtips (main)

feat/venue-proc-validation

0 0 0

Drew Skwiers-Koballa (47 minutes ago)

Ln 15, Col 31

Spaces: 4

UTF-8

LF

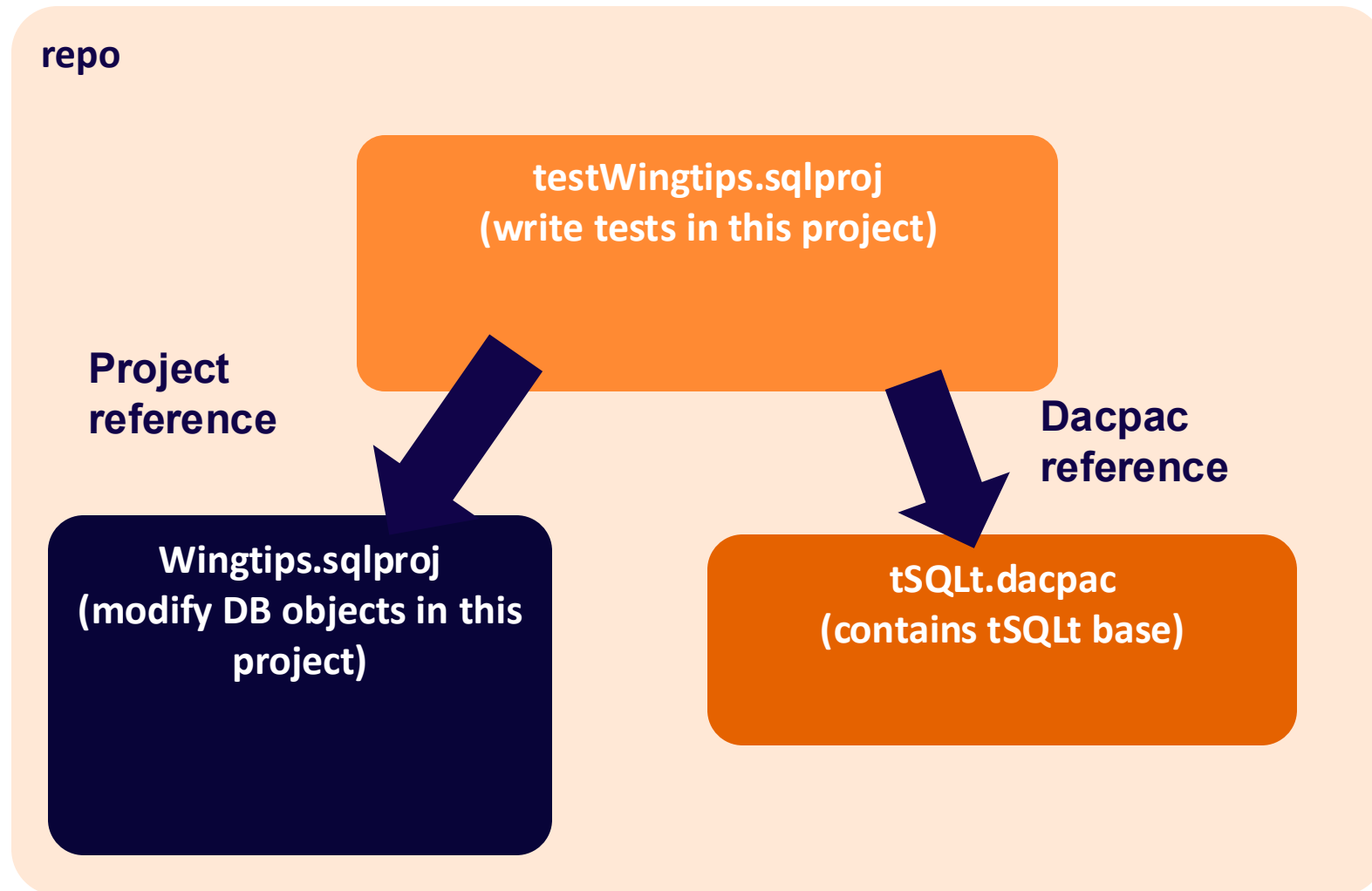
{ } Dockerfile

Unit testing – is it correct?

- Avoid unit tests on production environments
- Allows early detection of defects
- Increased confidence
- Test all the things
- Tests should be independent and small
- Automate as part of CI/CD

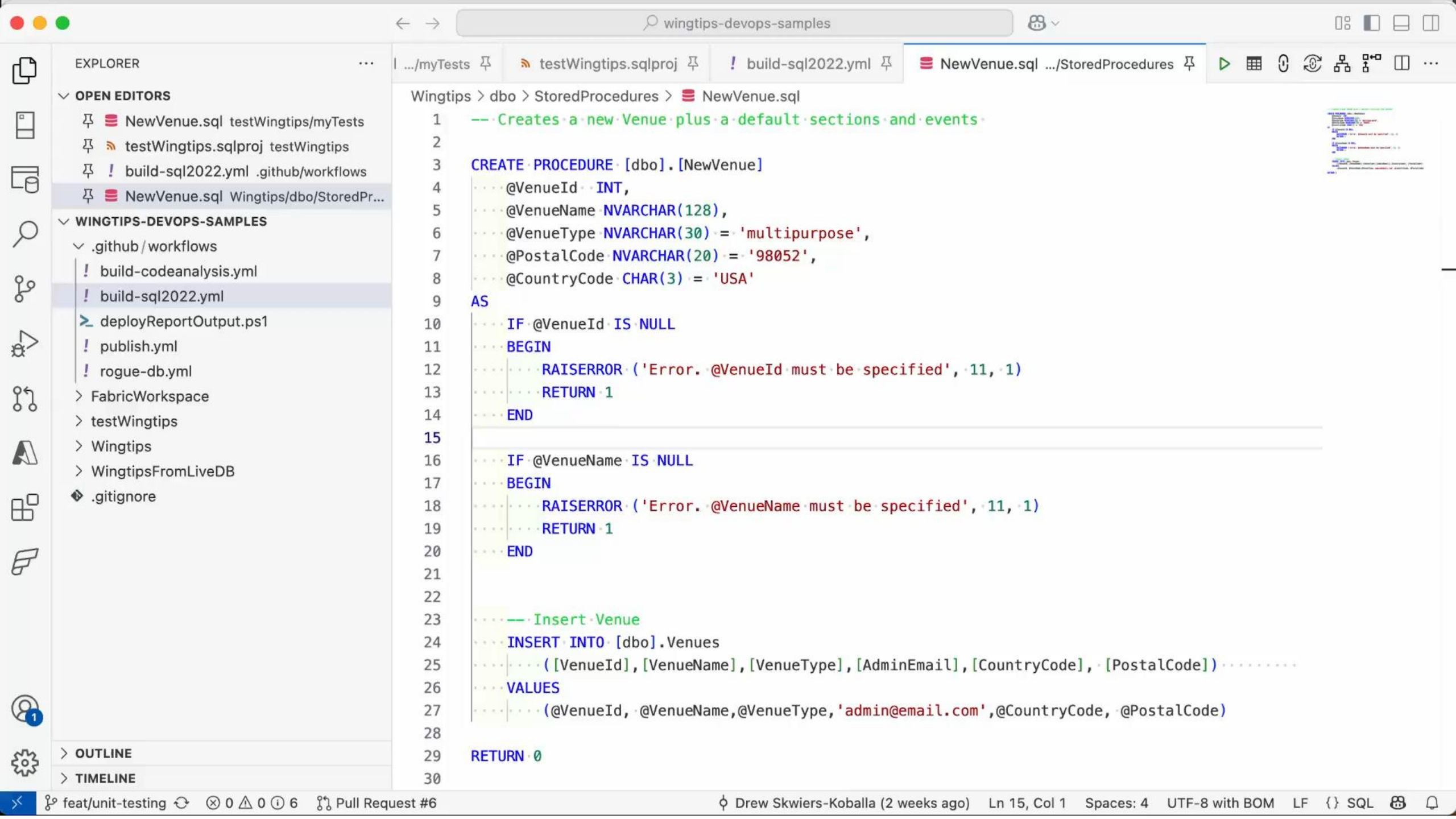


Writing, deploying, and running unit tests



Deploy testWingtips to a test instance with *IncludeCompositeObjects* property set to true

Use sqlcmd to execute utility scripts



EXPLORER

OPEN EDITORS

NewVenue.sql testWingtips/myTests
testWingtips.sqlproj testWingtips
build-sql2022.yml .github/workflows
NewVenue.sql Wingtips/dbo/StoredPr...

WINGTIPS-DEVOPS-SAMPLES

.github/workflows
build-codeanalysis.yml
build-sql2022.yml
deployReportOutput.ps1
publish.yml
rogue-db.yml
FabricWorkspace
testWingtips
Wingtips
WingtipsFromLiveDB
.gitignore

OUTLINE

TIMELINE

Wingtips > dbo > StoredProcedures > NewVenue.sql

```
1  -- Creates a new Venue plus a default sections and events
2
3  CREATE PROCEDURE [dbo].[NewVenue]
4  ... @VenueId INT,
5  ... @VenueName NVARCHAR(128),
6  ... @VenueType NVARCHAR(30) = 'multipurpose',
7  ... @PostalCode NVARCHAR(20) = '98052',
8  ... @CountryCode CHAR(3) = 'USA'
9  AS
10 ... IF @VenueId IS NULL
11 ... BEGIN
12 ...     RAISERROR ('Error. @VenueId must be specified', 11, 1)
13 ...     RETURN 1
14 ... END
15
16 ... IF @VenueName IS NULL
17 ... BEGIN
18 ...     RAISERROR ('Error. @VenueName must be specified', 11, 1)
19 ...     RETURN 1
20 ... END
21
22
23 ... -- Insert Venue
24 ... INSERT INTO [dbo].[Venues]
25 ... ([VenueId], [VenueName], [VenueType], [AdminEmail], [CountryCode], [PostalCode])
26 ... VALUES
27 ... (@VenueId, @VenueName, @VenueType, 'admin@email.com', @CountryCode, @PostalCode)
28
29 RETURN 0
30
```

Lab Three

- Static data via post-deployment script
- Deployment approvals
- Deploy script/report

Troubleshooting and optimising

- ...so, you know enough to be dangerous.

Pipeline best practices

- Specify environment version (avoid “windows-latest” and “ubuntu-latest”)
- If using a task, specify a version or a git commit hash (avoid latest)
- Watch for case sensitivity



My build is slow – what now?!

Unresolved references

- Completely unresolved references and case-mismatch delays the build
- Code sanitization takes time but pays off

Build fewer objects

- Project references extend the build time significantly
- Artifact/package references have much smaller impact
- Use project references only when contained objects change frequently and on same cadence

Complain

- No, seriously
- If you can share your project with Microsoft, reach out to

github.com/microsoft/dacfx

Branch management

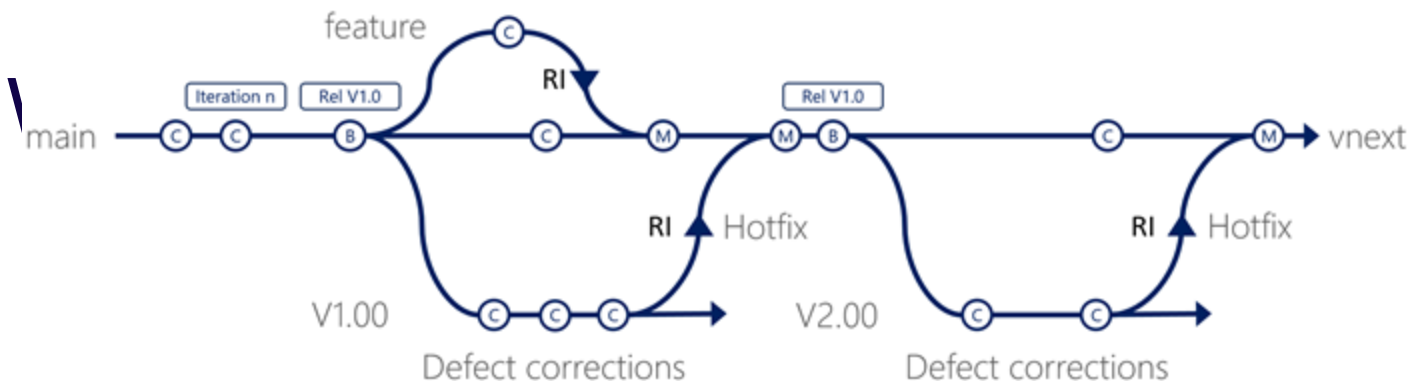
- Start simple 'main' branch



- Progressive branching



- Strategies



Tasks vs script directly

Script

- Script steps offer complete control
- Barebones operations
- .NET, SqlPackage, sqlcmd, and any CLI can be used directly
- Bash, cmd, PowerShell

GitHub azure/sql-action

- Windows+Linux
- Azure + anywhere
- Adds/removes firewall rule
- Builds a sqlproj, deploys a dacpac + other SqlPackage commands, runs a

Contributors welcome

ADO

SqlAzureDacpacDeployment

- Windows only, Azure only
- Picks out the version of SqlPackage*
- Adds/removes firewall rule from Azure
- Deploys a dacpac +

 Upcoming  renovation

Lab Four

- If we have time - optional

Wrap Up

What did we not talk about today?

- Publish profiles
- Deployment contributors
- Writing custom code analysis rules
- Refactorlog/refactoring

Resources

Documentation

- [Aka.ms/sqlprojects](https://aka.ms/sqlprojects)

More Community Members

- Kevin Chant
- Erik Ejlskov Jensen
- Kevin Pereira
- Olivier Van Steenlandt
- Erin Dempster

ONE LAST THING

Email: hamish@makestuffgo.com
for a 30-minute FREE consultation

PT afternoons are best for me

Your feedback is important to us



Evaluate this session at:

passdatacommunitysummit.com/evaluations

Thank you

Message for the end of the presentation
goes here

Hamish Watson

